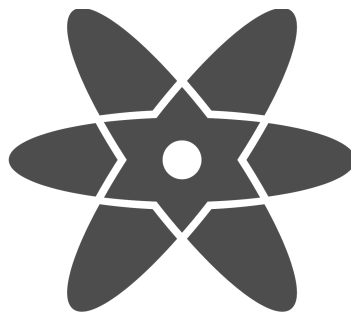




Xspress 3 X

User Manual

V1.2 2021

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1. Introduction

Xspress 3 X (X3X) is a fast readout system for X-ray fluorescence detectors. It is designed to operate at high input count rates with SDD or Ge detectors. X3X is available in 4 or more channel configurations and multiple units can be connected together to provide readout for larger detector arrays. Xspress 3 X has a user-friendly calibration procedure detailed in this manual.

In the box

Component

Xspress 3 X

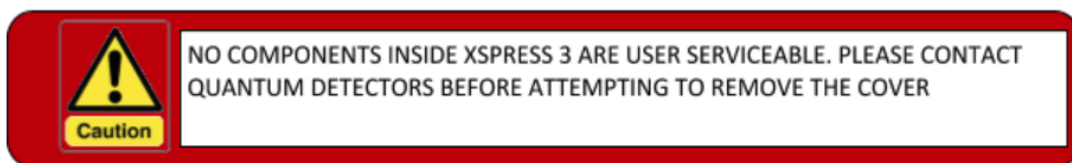
Control server

Power cables for X3X and server

Cat 7 network cable

Signal cables

OPTIONAL: Box to box single pole lemo cables for multibox systems
(>8 channels, see section 5)

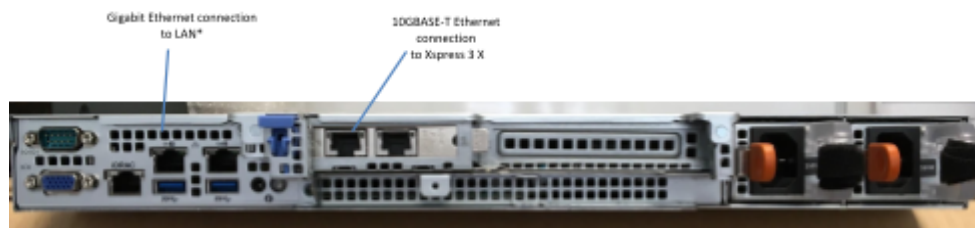


2. Configuring the control PC

Server configuration

This section details the setup procedure for the supplied control server. The server comes with a CentOS 7 Linux operating system and all the software you need to operate Xspress 3 X pre-installed.

Connect the Ethernet links as shown in the diagram below, connect X3X to the leftmost 10GBASE-T socket (not the onboard Gigabit LAN connection). If you have multiple X3X boxes as part of a larger system (>8 channels) connect boxes 2,3,4 to the remaining 10GBASE-T ports from left to right. For multibox X3X system setup, see section 5.



Rear panel of the supplied server

You may find the connection of a monitor and keyboard helpful for the initial setup. The server does have a Desktop environment installed but can also be run headless. Connect the AC power cables and turn the box on. You will be asked to log in. The default username is `xspres3` and the default password is `x5pr3553` (both are case sensitive). We recommend that you change the password to something more secure once you're set up.

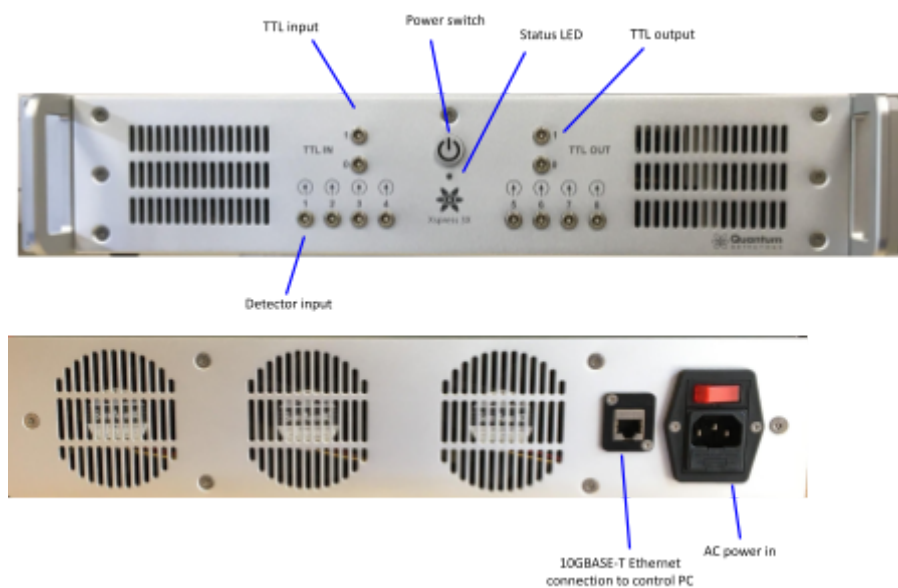
The primary Ethernet connection is set up as DHCP and will accept an IP address from your DHCP server. If you need to specify a static IP address then you can edit the file `/etc/sysconfig/network-scripts/ifcfg-em1`.

The file `/etc/sysconfig/network-scripts/ifcfg-p1p1` is configured with a static IP address for the private network that drives Xspress 3 X. This file should not be edited.

3. Configuring Xspress 3 X

Introduction

For synchrotron users, we recommend placing Xspress 3 X inside the hutch and keeping the signal cable from detector to electronics as short as possible. The signal cable we supply is 3m in length. The signal cable should be routed so as to avoid noise pickup from external sources (e.g. pumps, motors). Xspress 3 X should be powered from a clean mains circuit which is also free from these noise sources.



Front and rear panels of the Xspress 3 X

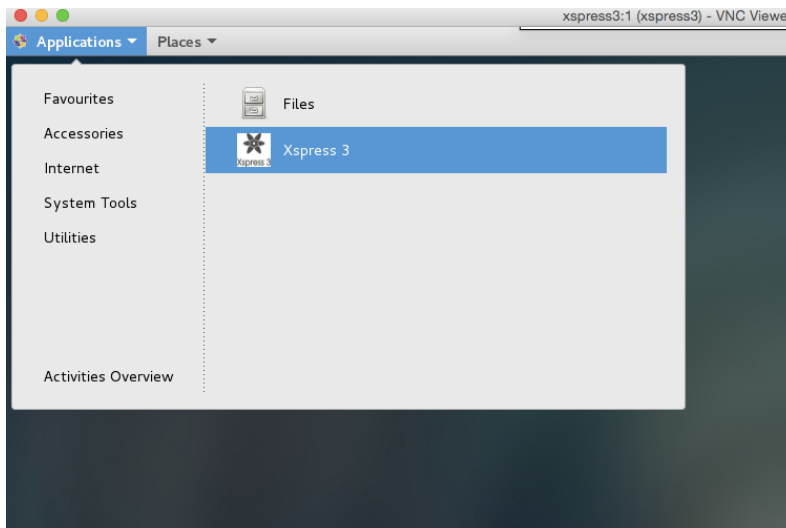
TTL input	5V TTL. TTL 1 is used in normal operation for an external gate signal. TTL 0 can be enabled by factory preset.
TTL output	5V TTL. Can be enabled by factory preset.
Detector input	Receives analogue signal from the detector.
Status LED	Amber when booting, green when ready.
Ethernet	Connect to control PC. Must be on private network.

Starting the calibration software

Xspress 3 X must be connected to the control server via a 10gb Cat 7 Ethernet cable on a private network. You may connect Xspress 3 X to the control PC through a direct cable or a patch panel, but not through a switch on a network. Xspress 3 X expects the control PC NIC to have the IP address 192.168.0.1.

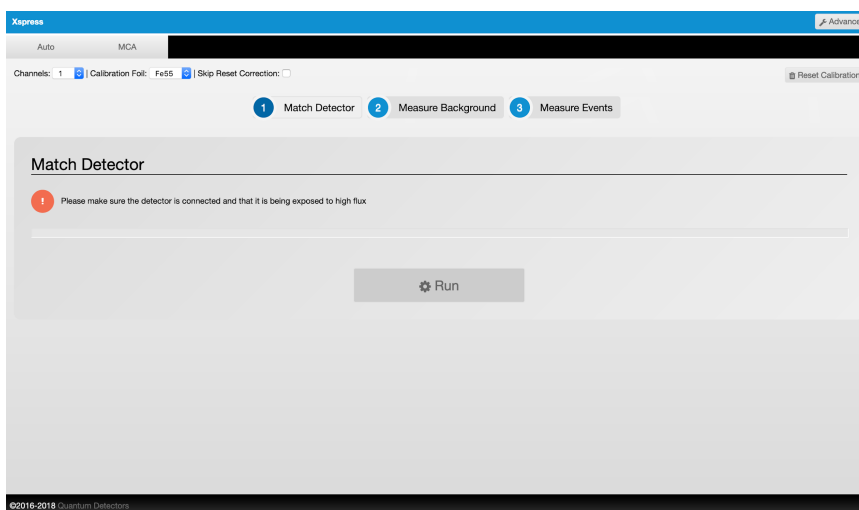
Quantum Detectors control server

- Connect Xspress 3 X to the control server
- Plug the power cord into the back of Xspress 3 X
- Wait for the LED on the front of Xspress 3 X to turn green
- Click on the 'Applications' button, then select 'Accessories' > 'Xspress 3'



Click 'Applications' > 'Accessories' > 'Xspress 3' to start the calibration software.

The calibration software will launch in a new terminal and web browser window.



Home screen for the calibration software.

4. Calibrating Xspress 3 X

Xspress 3 X must be calibrated with each detector it is intended to be used with. Once calibrated with a particular detector the calibration will remain stable indefinitely, with recalibration only necessary if the detector is repaired or its behaviour changes. Calibration can be completed in most situations via a simple 3-step automatic calibration procedure. This automatic procedure relies on the following conditions to be met:

- A reference sample target, of an element listed in the table [below](#)
- A monochromatic x-ray source
- Typical XRF experimental geometry (detector perpendicular to the beam)
- Low EM noise environment

If these conditions are not met then it may be necessary to use the advanced calibration procedure, details of which are explained later in this manual.

These conditions are stipulated to reduce the impact of complicating factors preventing the auto-calibration routine from accurately and dependably characterising the detector.

The list of reference sample targets has been chosen from elements that exhibit well defined and suitably isolated fluorescence peaks.

A monochromatic x-ray source, tuned to be above the target reference element fluorescence peak, eliminates the signal from fluorescence peaks at higher energies. This increases the available statistics at the target energy and reduces the potential of other peaks to confuse the automated routine.

Placing the detector perpendicular to the beam minimises the amount of scatter from the target sample.

A low EM noise environment reduces the likelihood of external noise degrading the signal from the detector. Care should be taken to route the signal cables away from any interference sources.

Element	Emission line used as reference	Emission line energy (keV)
Mn (also ⁵⁵ Fe)	K	5.895
Fe	K	6.400
Pt	L	9.434
Mo	K	17.443
Br	K	11.908
Cu	K	8.042
Au	L	9.704
Zr	K	15.746
Pd	K	21.123
Sb	K	26.272
Se	K	11.208

List of target elements supported by the automatic calibration procedure.

Calibration via the auto-calibration tool

Before you calibrate the detector, hit the 'Reset Calibration' button on the top right of the calibration screen. This ensures you are starting afresh, with no settings lingering from a previous calibration.

Select the appropriate number of input channels in the top left of the interface.

Make sure that your detector output(s) are connected to the signal input(s) on the front panel of your Xspress 3 X system.

Ensure the detector is ready to collect data, e.g. by checking any status indicators.

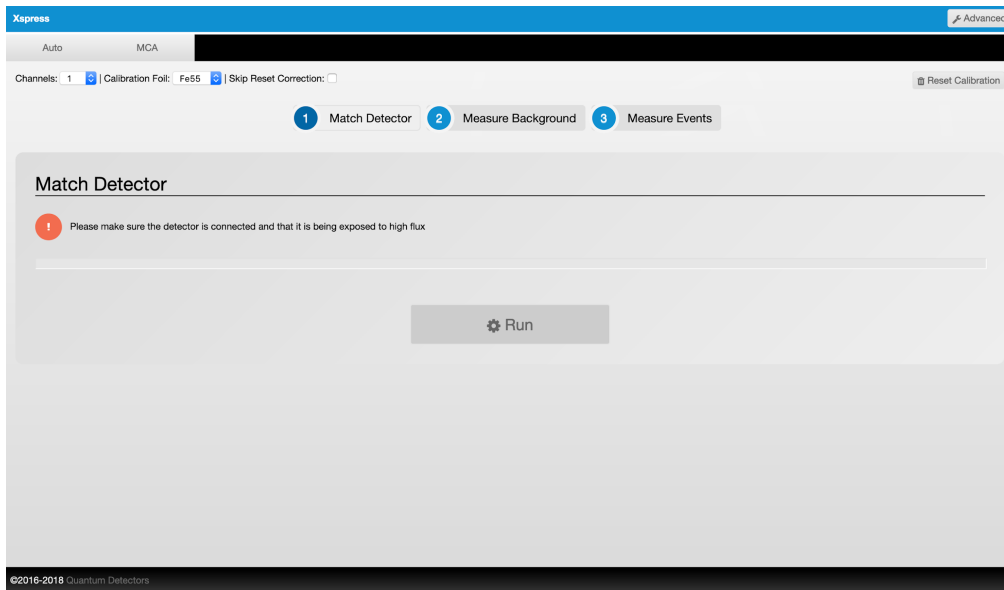
Choose an appropriate radioactive source or sample foil to calibrate Xspress 3 X. For the automatic calibration to succeed (or to simplify manual calibration) it is important that the source or foil generates a simple spectrum. A list of supported elements for auto calibration can be seen by clicking on the 'Calibration Foil' drop down in the top left of the screen. Supported elements are listed in the table on the next page.

The automatic procedure has three main steps. The first aims to match the analogue input range of Xspress 3 X to the analogue output range of your detector; the second measures the background signal from the detector; and the third measures the shape and nature of events.

Step 1: Match Detector

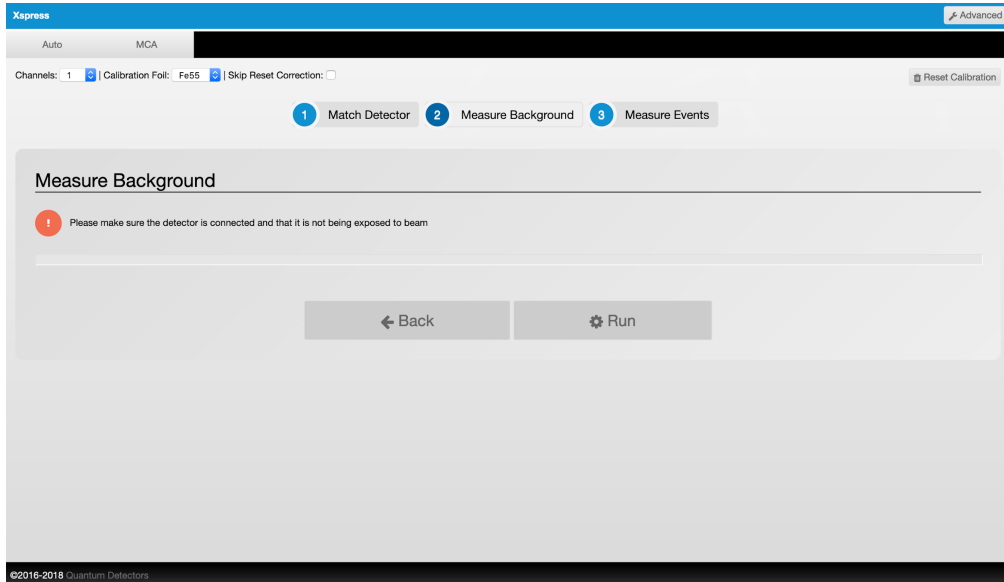
The detector preamp will output a given voltage range over its reset cycle. Xspress 3 X will match this range to ensure that data is not missed and that the signal to noise ratio is optimised.

Ensure that the detector is being exposed to photons at a rate of approximately 20-30 kHz. Click on 'Match Detector' and click 'Run'. When complete, the message 'Calibration Step Completed' will appear.



Step 2: Measure Background

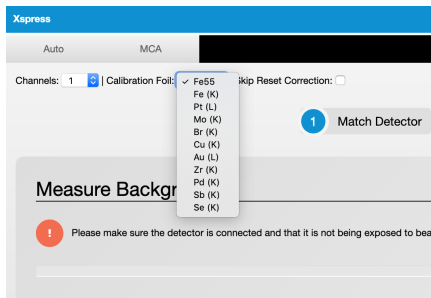
Next, click on 'Measure Background'. Cover the source or close the beam shutter so that no photons reach the detector. This step allows Xspress 3 X to measure the background signal from your detector, which is important as it gives us a baseline from which to measure events. Once the source is covered, click on 'Run'.



Step 3: Measure Events

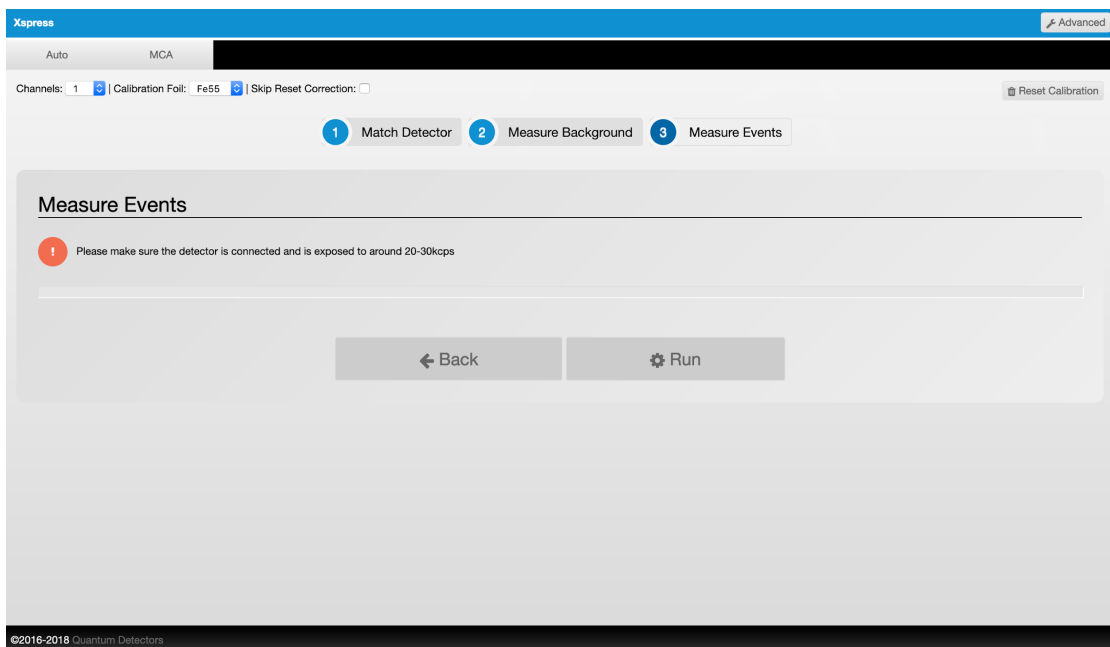
Finally, Xspress 3 X needs to learn the shape of the events, account for the detector reset, and perform an energy calibration. Select the element you are using to

calibrate with from the 'Calibration Foil' drop down menu in the top left of the browser window.

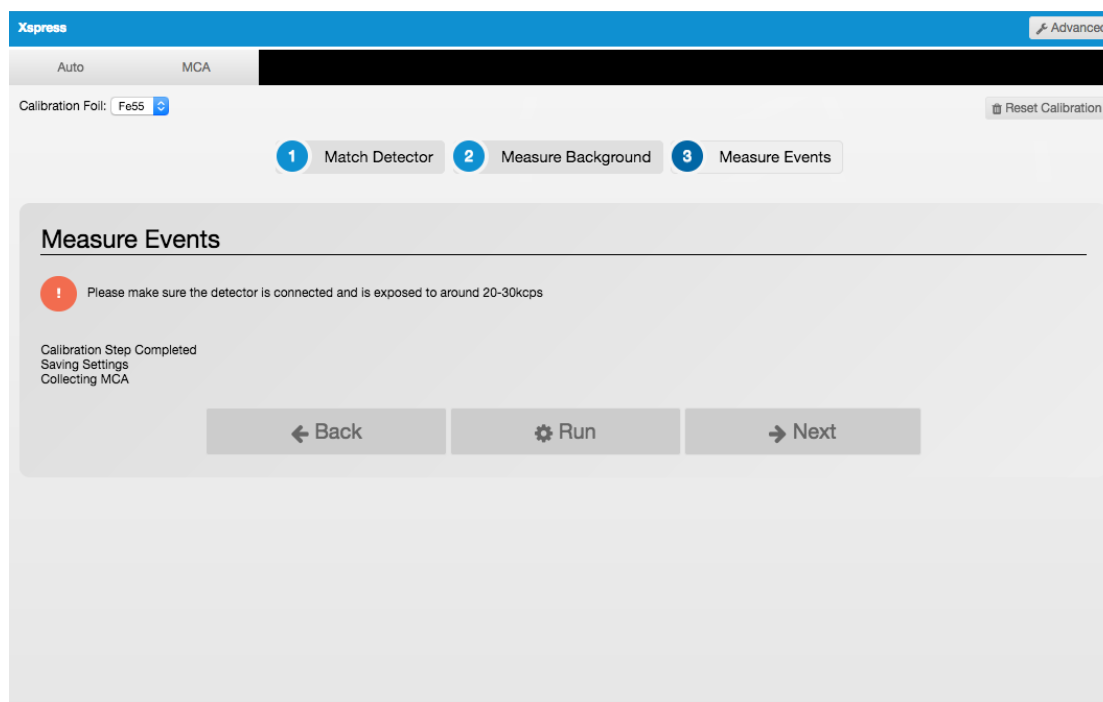


Note: if you're using a Manganese foil, select Fe55.

Uncover the source or open the beam shutter and then click on 'Measure Events'. There should be a rate of 20-30 kHz incident on the detector. Click 'Run'.

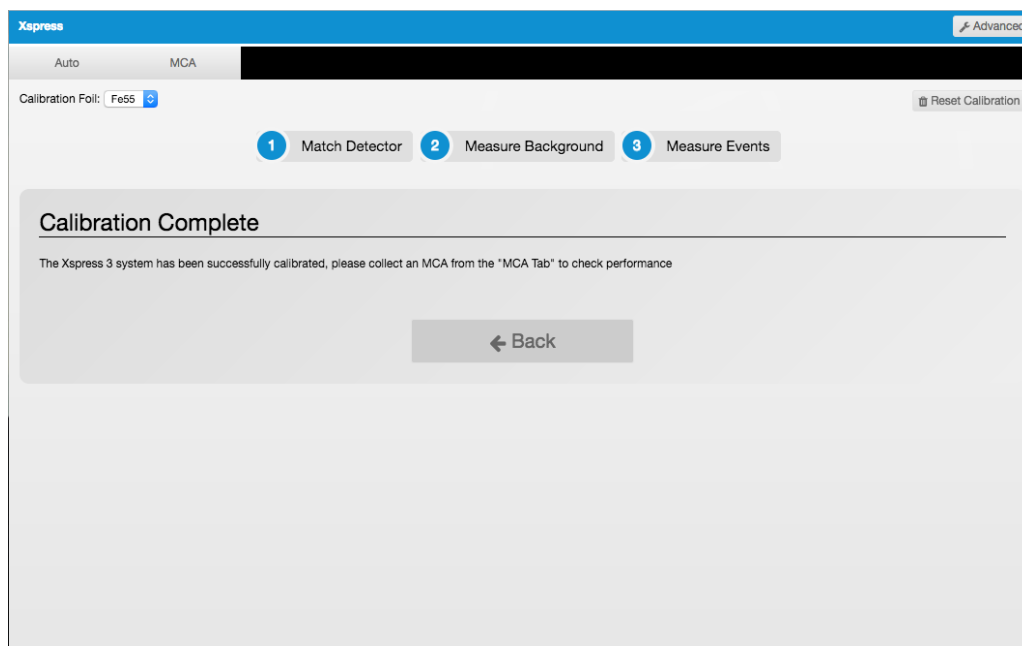


Once complete you'll see the message 'Calibration Step Completed'

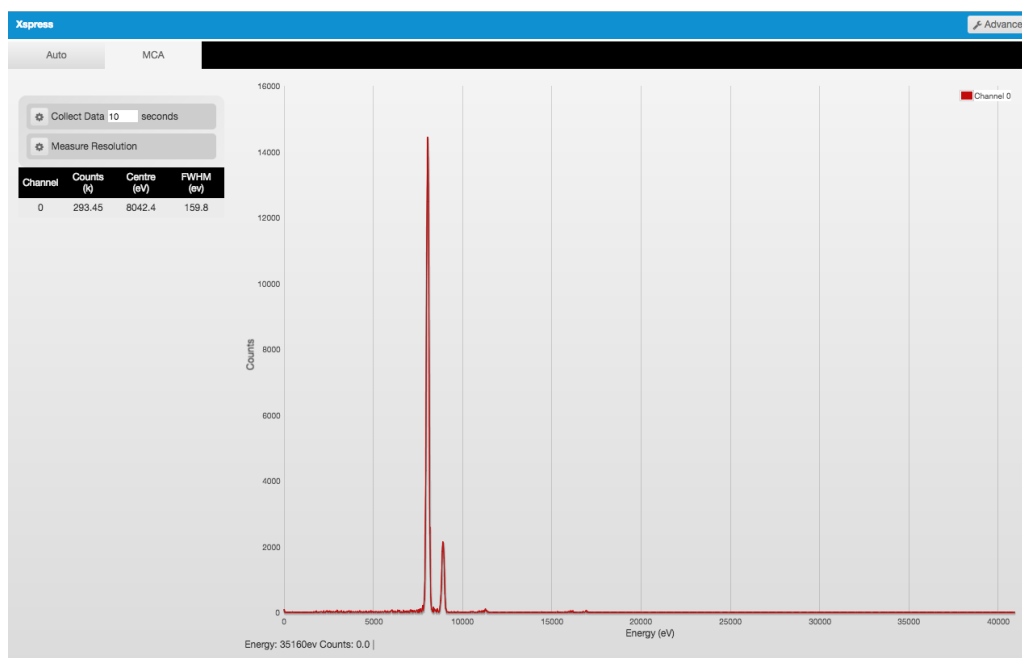


The measure events step can take a few minutes. As part of this process, Xpress 3 searches for detector resets and attempts to correct the tail at the end of a reset via an iterative process of curve fitting. In some circumstances, this process can get stuck and the calibration appears to stall for several minutes. If this happens, you may wish to skip the reset tail correction. Check the box marked 'Skip Reset Correction' at the top of the screen and run the 'Measure Events' step again.

Click on 'Next', you'll be prompted to collect an MCA to check the energy calibration and peak resolution.



Click on the 'MCA' tab, then click on the cog next to 'Collect Data'. 10 seconds is the default time, if you wish to collect for a longer or shorter period the preferred value can be entered into the box. The maximum collection time is 50 seconds.



Check that the peak is in the right place for the sample that you used to calibrate and that the energy resolution is satisfactory¹.

¹ See <https://quantumdetectors.github.io/xray-utils/> for a complete online emission line reference tool.

Calibration via the advanced mode

Clicking on the 'Advanced' button in the top right of the user interface will reveal extra tools that may be used to complete a full calibration of a detector or to fine tune a calibration made previously.

There are seven steps to the advanced calibration procedure, each of which should be followed in the order given below:

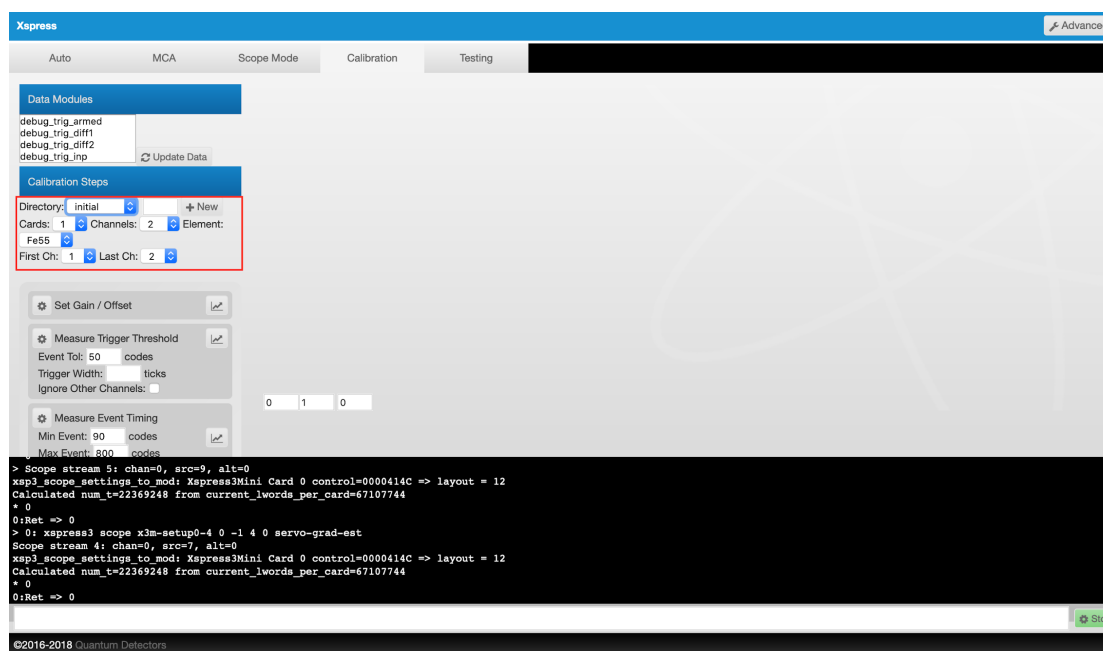
- Initial settings
 - Setting the calibration folder
 - Setting the target element
 - Channel number and channel specification
- Set Gain / Offset
 - Matching the input gain to the detector
- Measure Trigger Threshold
 - Measure the background
- Measure Event Timing
 - Categorise the event rise time
- Measure Event Shape
 - Categorise the shape of events
- Measure Reset Tail
 - Categorise the nature of the detector reset
- Measure Event Gain
 - Calibrate the energy of the events

All steps, with the exception of Measure Trigger Threshold, should be run with an x-ray flux of approximately 20-30 kHz incident on each element of the detector(s).

The Set Gain section is analogous to the Match Detector page in the Auto mode, the Measure Trigger Threshold section is analogous to the Measure Background tab in the Auto mode and the rest of the sections are completed by the final stage of the Auto mode.

If you use the Advanced mode to run through the calibration procedure, be sure to reset the calibration first, otherwise you can find yourself with conflicts in the calibration files, leading to poor calibrations. You should then run through the items, from top to bottom, in the order specified. If you need to return to an earlier step then you should re-run each subsequent step.

Initial settings



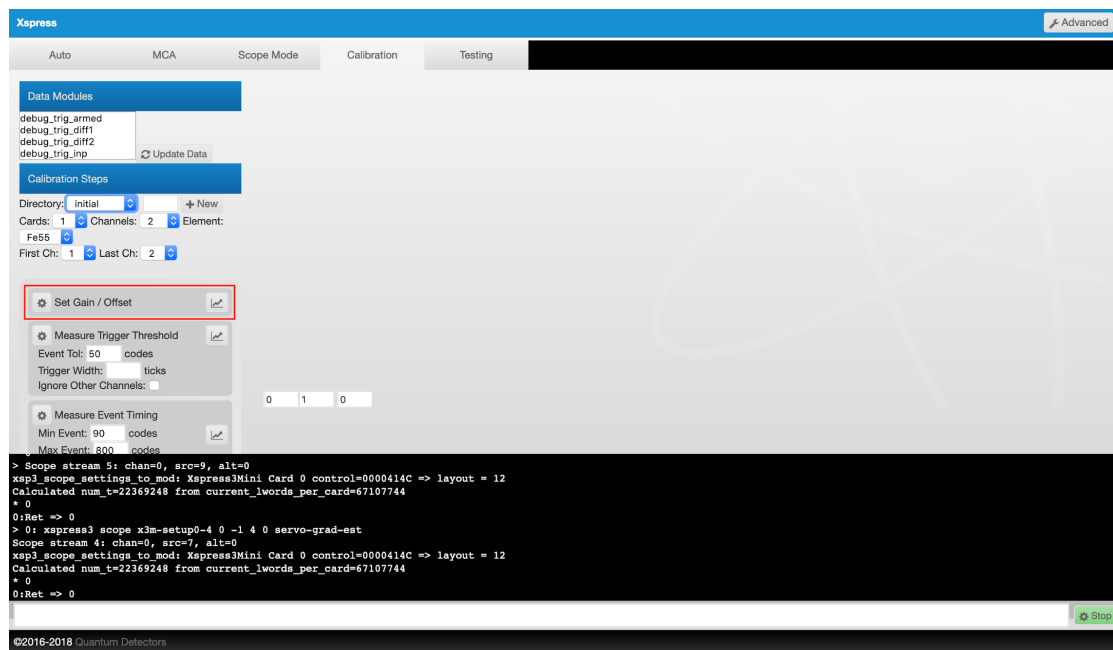
In this step one can choose the name of the directory to save the calibration files. When Xspress 3 X is first started, the system defaults to the 'initial' folder. It is advised to use this folder when calibrating a detector. Once a calibration is completed then the initial folder can be copied to create a folder with a more descriptive name.

The 'Cards' and 'Channels' values are automatically set when the calibration software is loaded. There are two channels per card on the X3X system. Thus, a four channel system will have two cards, 8 channels 4 cards etc.

One can calibrate channels independently by selecting the 'First Ch' and 'Last Ch' appropriately.

Select the element to match the target element or source you intend to use. If you decide, or are obliged, to use a target not in this list then you will need to enter a custom peak energy at the event gain stage, later in this process.

Set Gain / Offset

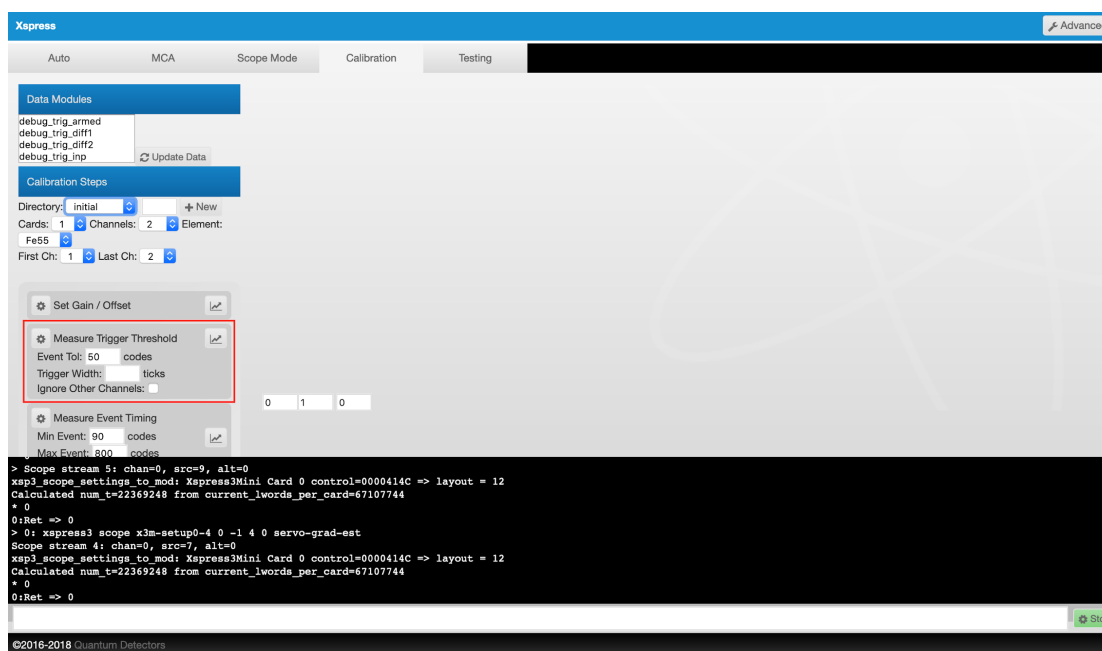


This step matches the gain of the X input stage to the range of the detector. It must be completed with flux incident on the detector. Click the cog to the left of the 'Set Gain / Offset' text to begin the process.

On multi-channel detectors, where the resets are set to be simultaneous across the channels to prevent cross-talk, a primary channel may prevent the secondary channels from moving through their full range before being forced to reset.

Once completed the software will display the ranges calculated for each channel. Check the ranges match what you expect of your detector(s) and consider editing them if they differ.

Measure Trigger Threshold

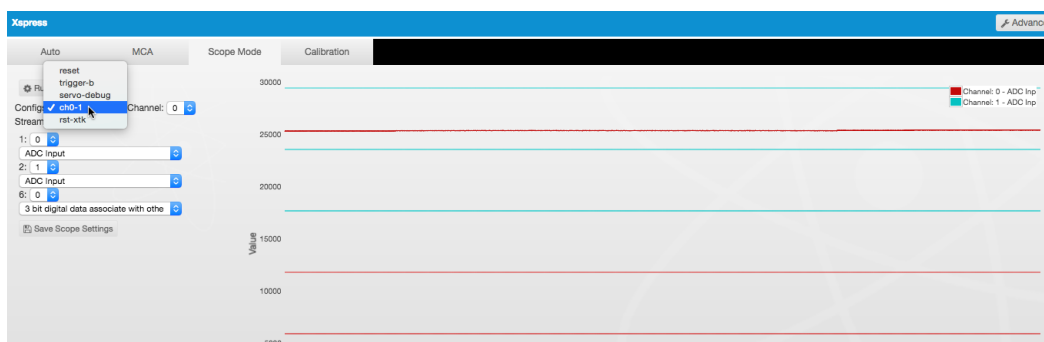


This stage is used to measure the background of the detector. Ensure no x-rays are incident on the detector and then click the cog symbol to begin measurement.

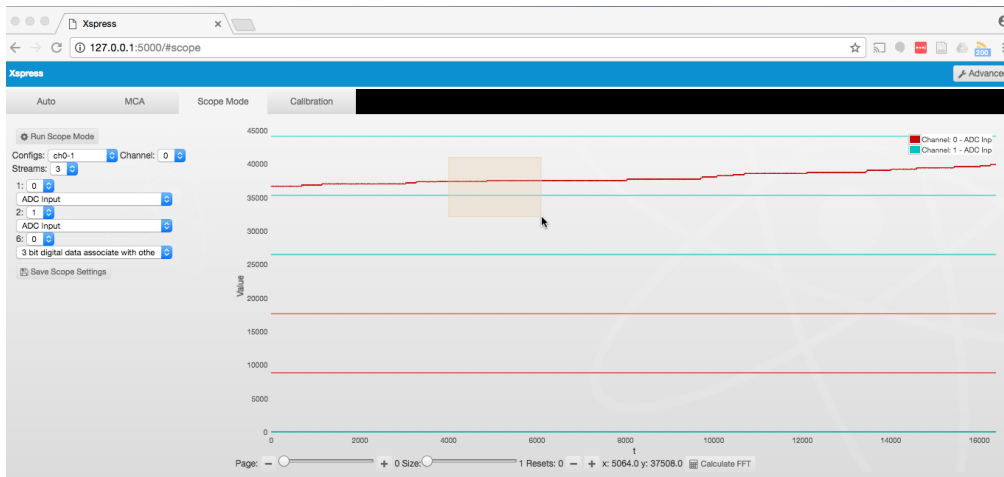
Low energy tailoring

Users who intend to use their system in the low energy regime (less than 2 keV) should conduct the following process to tailor the trigger threshold. The size of low energy events tends to be very close to the size of noise events and so one must inspect the noise more carefully.

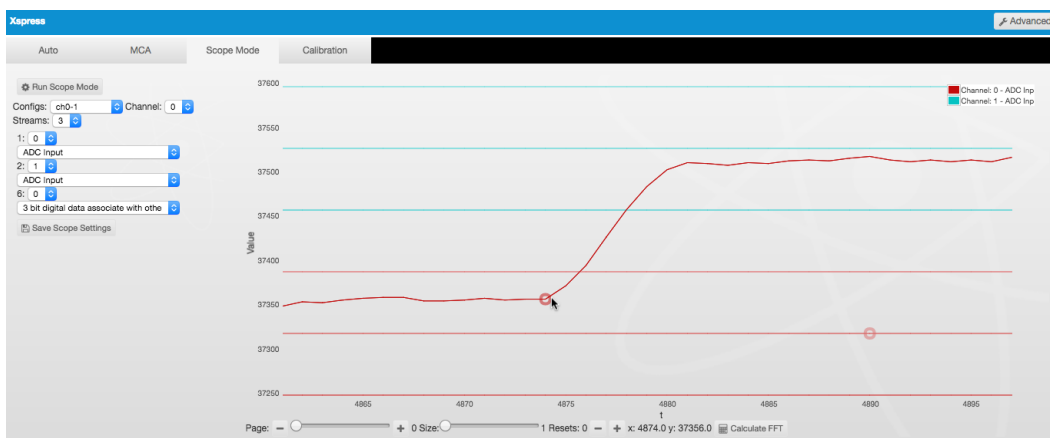
Click on the 'Scope Mode' tab and select 'ch0-1' from the 'Configs' dropdown.



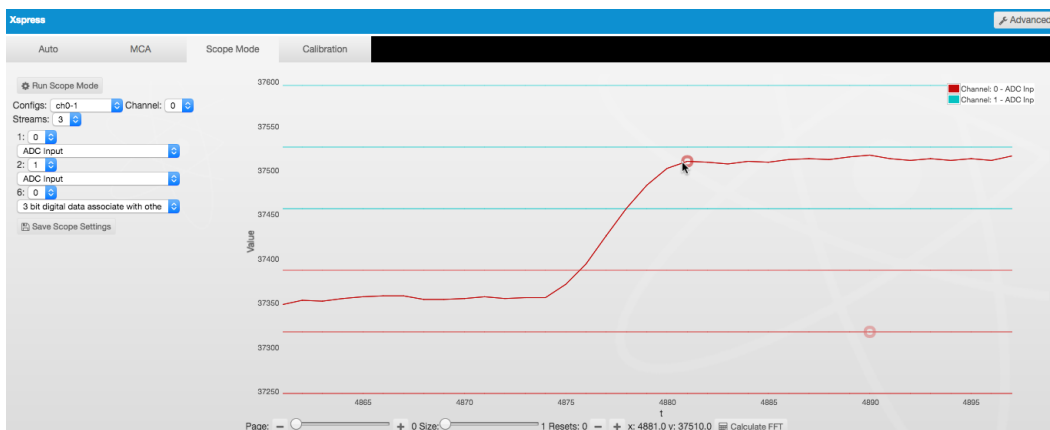
Click the 'Run Scope Mode' button and wait until you see a trace with events displayed in the GUI. Find an event and zoom in on it by dragging a box around the event.



Place your cursor at the start of the event and take a note of the 'x' position displayed at the bottom of the Scope page.



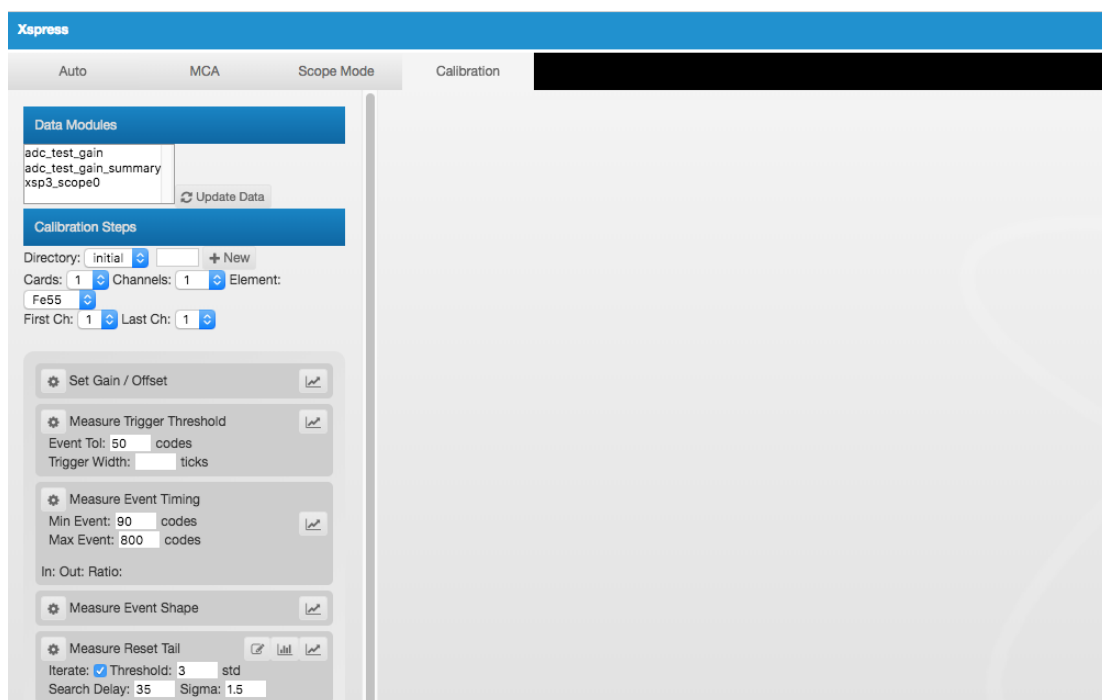
Place your cursor at the end of the event and take a note of the 'x' position displayed at the bottom of the Scope page.



Subtract the start 'x' position from the end 'x' position and make a note of that number. In this case, $4881 - 4874$ gives 7 clock ticks. This is the trigger width, and represents the rise time of an event in Xspress 3 clock cycles.

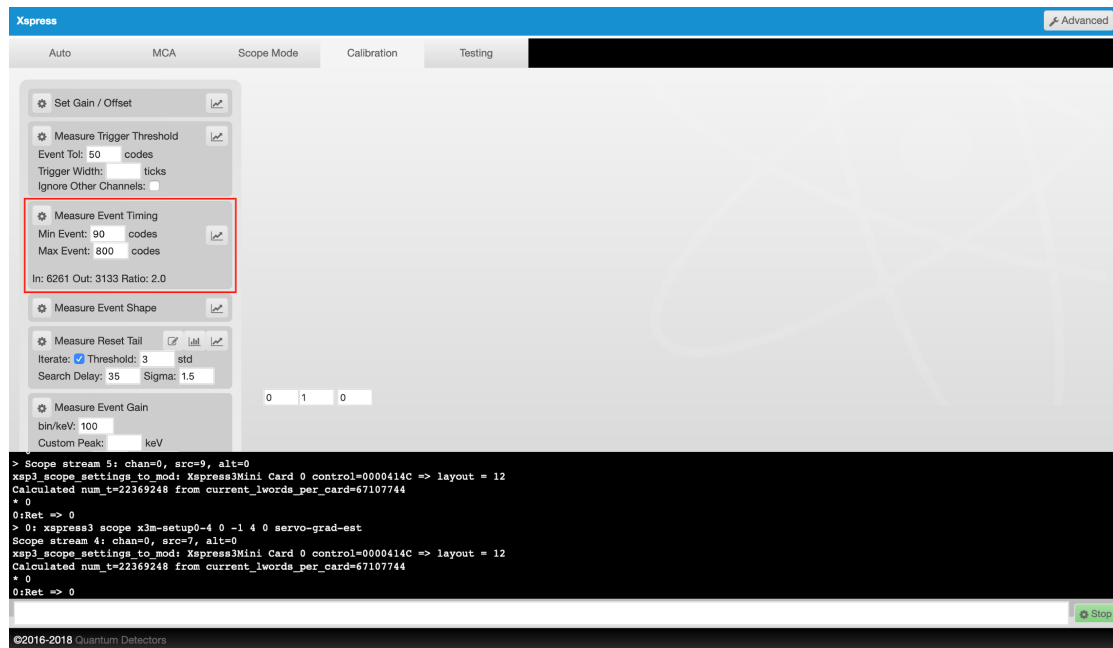
Take a note of the height of the event by subtracting the corresponding the 'y' figures. In this case it is approximately 140 ADC codes, though it may be smaller on your detector.

Close the beam shutter so that no photons reach the detector. Click on the 'Calibration' tab and enter the 'Trigger Width' into the box under 'Measure Trigger Threshold'. Ensure that the 'Event Tol' is smaller than the size of the event you measured, but bigger than the average range of the noise. Click the cog to begin the trigger threshold measurement process.



Xpress 3 will now measure the trigger thresholds twice, and automatically adjust the event filters for low energy events.

Measure Event Timing



This stage measures the rise time of events to enable the Xspress system to better identify true events over noise events or resets.

Open the shutter so that photons reach the detector at a rate of approximately 20-30 kHz. If you are calibrating at low energy then make sure that the number in the 'Min Event' box is lower than the event height you measured earlier. Otherwise, follow the process described in the section [Low energy tailoring](#) (above) to measure the typical size of an event. Once you have this measure, ensure that the 'Min Event' and 'Max Event' values are set so that a typical event is detected.

Click on the cog next to 'Measure Event Timing' to begin measurement.

If the ratio reported is greater than 3 you must reduce the rate of incident x-rays and run again.

Measure Event Shape

The screenshot shows the Xpress software interface with the 'Testing' tab selected. The 'Measure Event Shape' panel is highlighted with a red box. The panel contains the following settings:

- Set Gain / Offset**: [Icon]
- Measure Trigger Threshold**: [Icon]
 - Event Tol: 50 codes
 - Trigger Width: [] ticks
 - Ignore Other Channels: ☐
- Measure Event Timing**: [Icon]
 - Min Event: 90 codes
 - Max Event: 800 codes
 - In: 6261 Out: 3133 Ratio: 2.0
- Measure Event Shape**: [Icon] (highlighted with a red box)
- Measure Reset Tail**: [Icon]
 - Iterate: ☒ Threshold: 3 std
 - Search Delay: 35 Sigma: 1.5
- Measure Event Gain**: [Icon]
 - bin/keV: 100
 - Custom Peak: [] keV

Below the configuration panels, there is a terminal window showing the following output:

```
> Scope stream 5: chan=0, src=9, alt=0
xsp1_scope_settings_to_mod: Xpress3Mini Card 0 control=0000414C => layout = 12
Calculated num_t=22369248 from current_lwords_per_card=67107744
* 0
0:Ret => 0
> 0: xpress3 scope x3m-setup0-4 0 -1 4 0 servo-grad-est
Scope stream 4: chan=0, src=7, alt=0
xsp3_scope_settings_to_mod: Xpress3Mini Card 0 control=0000414C => layout = 12
Calculated num_t=22369248 from current_lwords_per_card=67107744
* 0
0:Ret => 0
```

At the bottom right, there is a 'Stop' button.

This stage measures the shape of events to characterise and reduce the effect of ringing in the signal after an event occurs.

Ensure that photons reach the detector at a rate of approximately 20-30 kHz. Click on the cog next to 'Measure Event Shape' and wait for it to complete.

Measure Reset Tail

The screenshot shows the Xpress software interface with the 'Testing' tab selected. The 'Measure Reset Tail' panel is highlighted with a red box. The panel contains the following settings:

- Measure Event Shape**: [Icon]
- Measure Reset Tail**: [Icon] (highlighted with a red box)
 - Iterate: ☒ Threshold: 3 std
 - Search Delay: 35 Sigma: 1.5
- Measure Event Gain**: [Icon]
 - bin/keV: 100
 - Custom Peak: [] keV
 - Bin Range: [] to []
- Save Settings**: [Icon]
- Create EPICS Links**: [Icon]
- Shorten Running Avg**: ☐

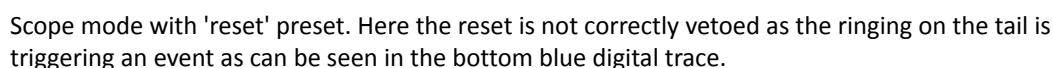
Below the configuration panels, there is a terminal window showing the following output:

```
> Scope stream 5: chan=0, src=9, alt=0
xsp1_scope_settings_to_mod: Xpress3Mini Card 0 control=0000414C => layout = 12
Calculated num_t=22369248 from current_lwords_per_card=67107744
* 0
0:Ret => 0
> 0: xpress3 scope x3m-setup0-4 0 -1 4 0 servo-grad-est
Scope stream 4: chan=0, src=7, alt=0
xsp3_scope_settings_to_mod: Xpress3Mini Card 0 control=0000414C => layout = 12
Calculated num_t=22369248 from current_lwords_per_card=67107744
* 0
0:Ret => 0
```

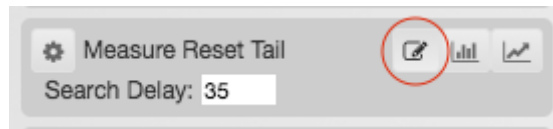
At the bottom right, there is a 'Stop' button.

This step can take a little while to complete, watch for the output in the console at the bottom of the window to stop updating. If it is still running after five minutes then it is likely the algorithm has been caught in a local minima and so will not successfully complete. In this case, refresh the browser and run the process again, this time with the 'Iterate' checkbox deselected.

When the detector resets, Xpress 3 X must detect this and veto any fluctuations in the detector signal until reset has finished. If the veto period is much longer or much shorter than the reset from the detector, one may see poor peak resolution or spurious peaks in the MCA.

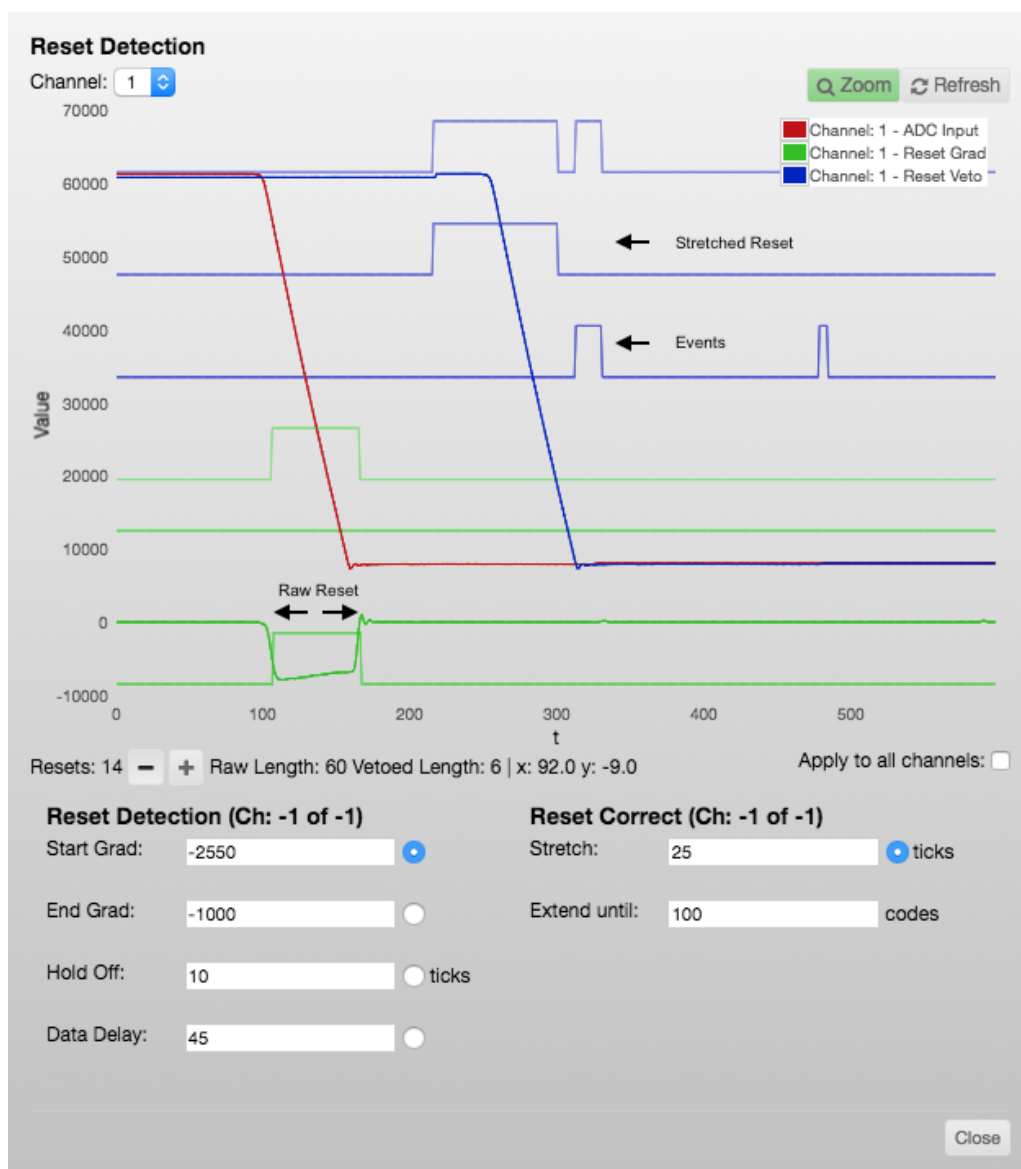


Ensure that the detector is receiving photons at a rate of 20-30 kHz. Click on the pencil icon to the right of the 'Measure Reset Tail' text.



This will bring up the Reset UI. The software will take a few seconds to collect the relevant data streams to analyse the reset detection and correction.

By default the 'Zoom' button is highlighted in green, this allows you to drag and zoom on the plots, double click to zoom out. Clicking again to disable zoom then allows you to capture values by dragging and clicking on the plot. Refresh collects a new series of scope data to allow changes to be visualised in real time.



There are two distinct parts to dealing with reset in Xspress 3 X: detecting the reset and correcting the reset.

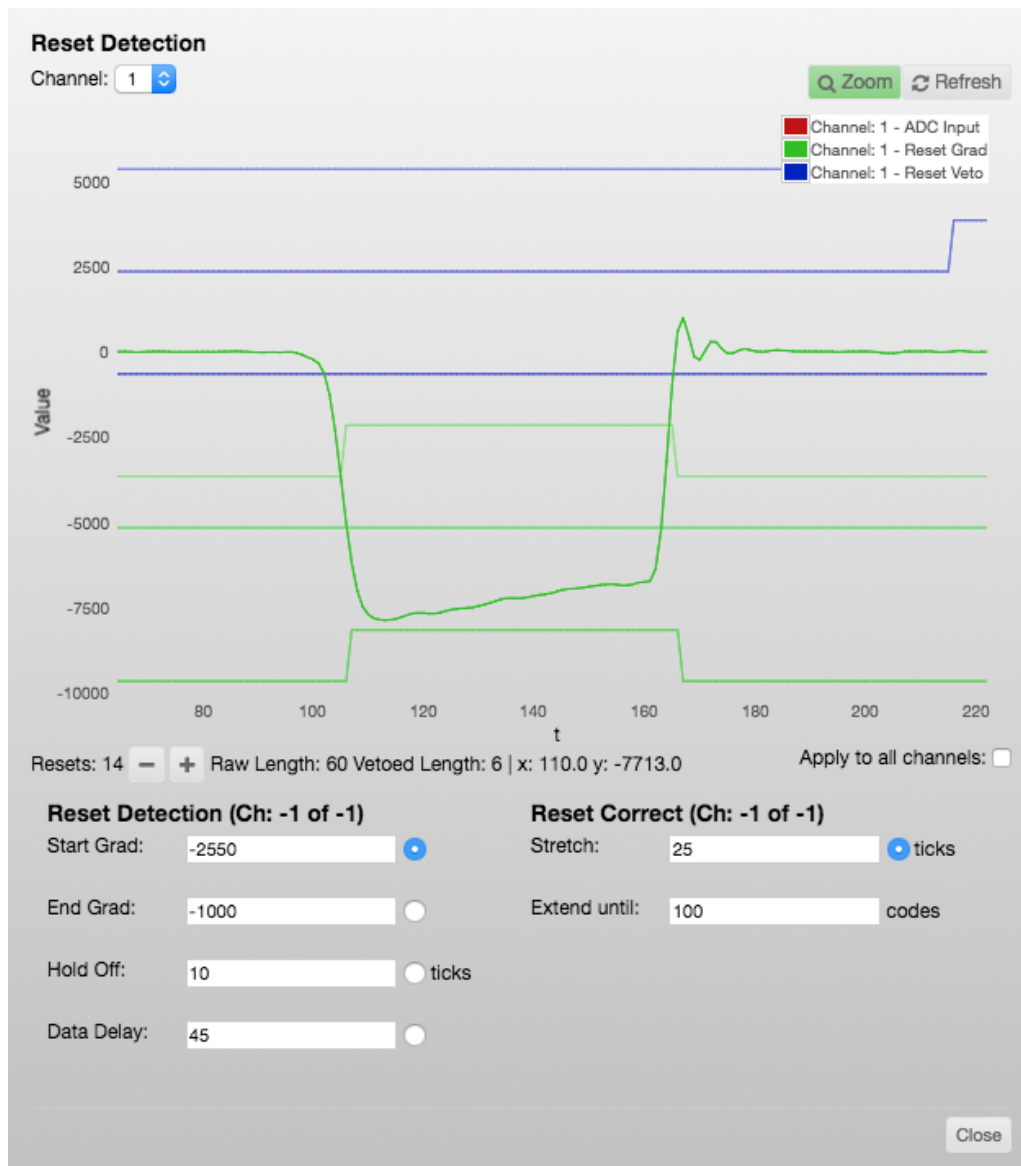
Detecting the reset is done by looking for changes in the gradient of the detector signal using the 'Start Grad' and 'End Grad' values in the 'Reset Detection' section of the UI. 'Data Delay' is used to offset the data stream from the event that triggers the reset. 'Hold Off' defines a period of time to prevent nuisance triggers around the reset.

The second part of the UI allows for configuration of the Reset Corrector, this extends the 'Raw Reset' by a number of ticks, and/or allows it to be extended until the ADC values comes back above the 'Extend Until' value. Be aware this is in ADC codes (the y-axis in the UI), not ticks (the x-axis in the UI).

The plot traces are as follows:

- The red trace is the raw ADC stream.
- The green trace is the gradient of the red trace, this gradient is used to trigger the reset start and end as defined in 'Start Grad' and 'End Grad'. The green digital streams show the 'Raw Reset Length', the UI also calculates the length of this value too, in this case it is 60 ticks.
- The blue trace is the output from servo, which can be used to infer the extended length of the reset. In this case the 'raw' reset value of 60 ticks has been extended by 25 ticks of the 'Stretch' time in Reset Correct. The digital streams are bottom to top, detected events, reset, events + reset.

Reset detection

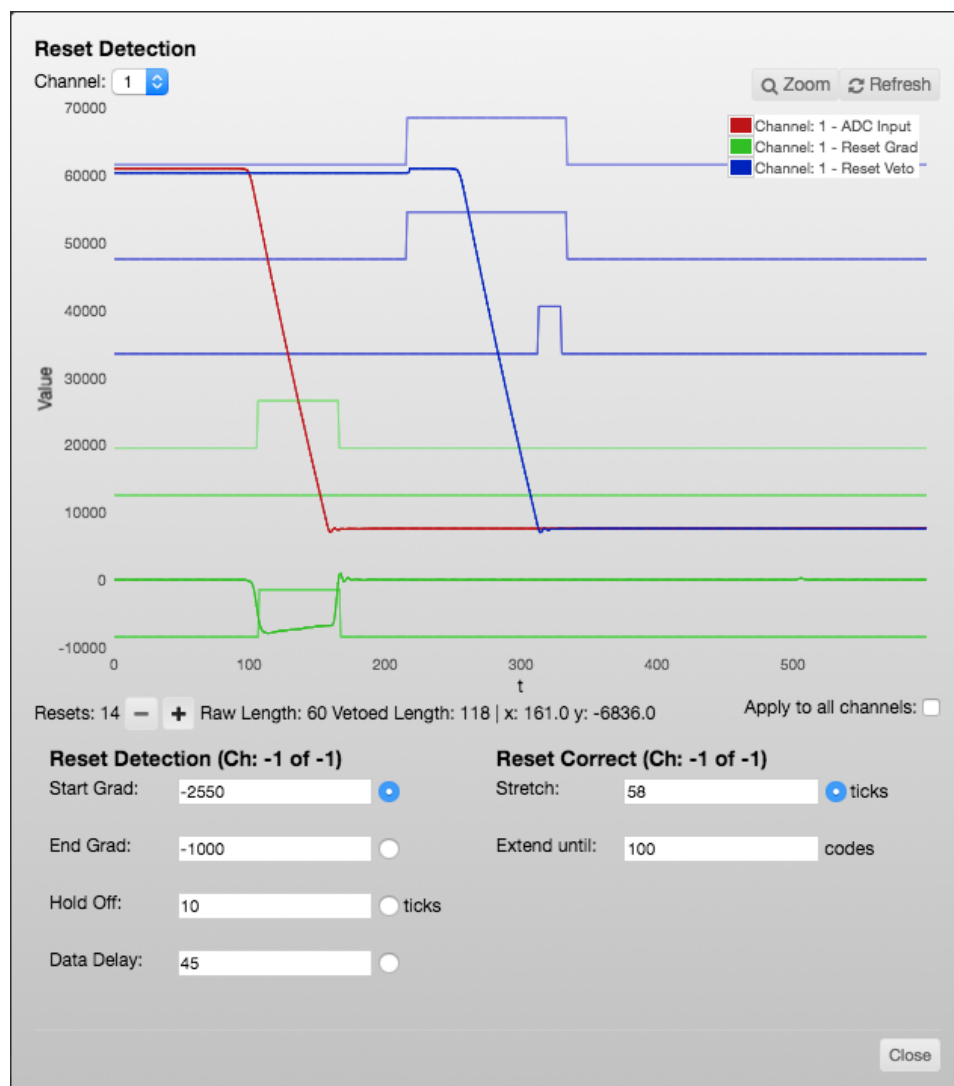


Above is a zoomed in view of the 'Reset Grad' trace. It can be clearly seen that the reset is triggered at the specified -2550 value, and finishes as the gradient passes through -1000. By clicking 'Zoom' again zooming will be disabled. Now clicking on the green trace at a specific point will capture that value into the field that has the blue radio box enabled. In this case it will be captured into the 'Start Grad' field. You can click the radio box next to 'End Grad' to capture a value into this field.

Reset Correction

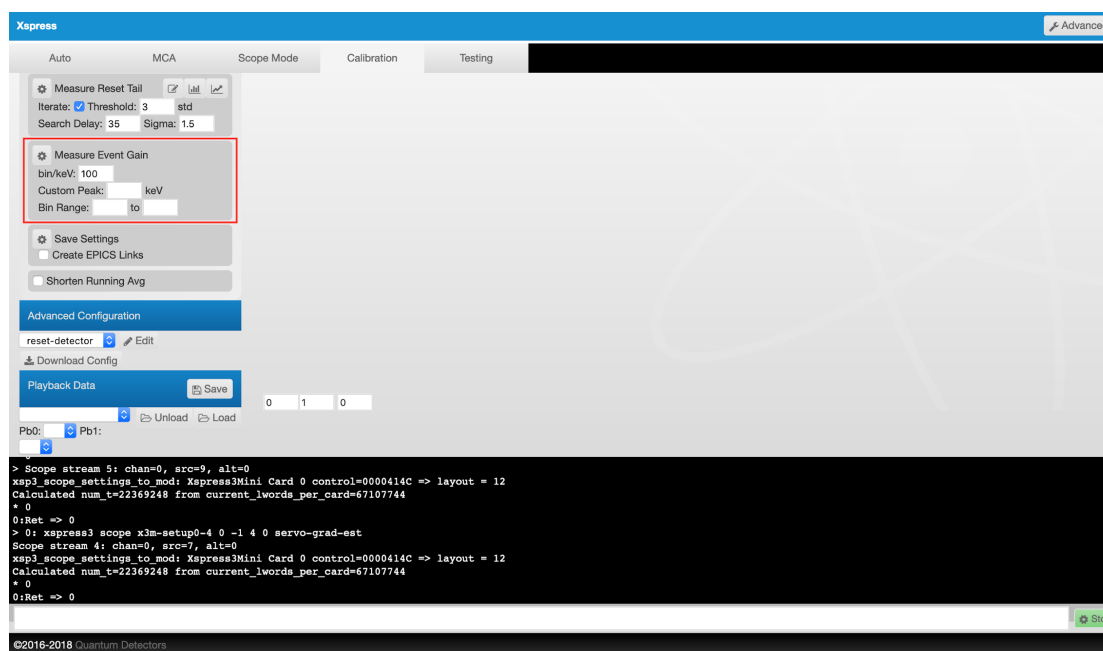
In the plot above one can see that Xpress 3 X has detected a spurious event caused by the ringing of the event tail. In this case the 'Stretch' time needs to be increased in order to veto the end of the reset tail ringing.

Make sure 'Zoom' is disabled (i.e. not green). Now drag along the blue trace from start to end of where the reset should be vetoed. The UI will calculate the 'Stretch' time from the dragged value minus the length of the raw reset. Then click 'Refresh' to see what effect the change had.



Once the UI has refreshed and new data has been collected, one can see 'Stretch' has now been updated to 58 ticks. The spurious event is now vetoed and the UI correctly reports the Vetoed Reset length as 118 ticks = Raw length of 60 + Stretch time of 58.

Measure Event Gain



The last step required is to perform an energy calibration. If you are using a standard target you can simply click the cog to run the scaling calibration with the default settings. Once it is complete, use the MCA tab to check on the result.

Custom scaling

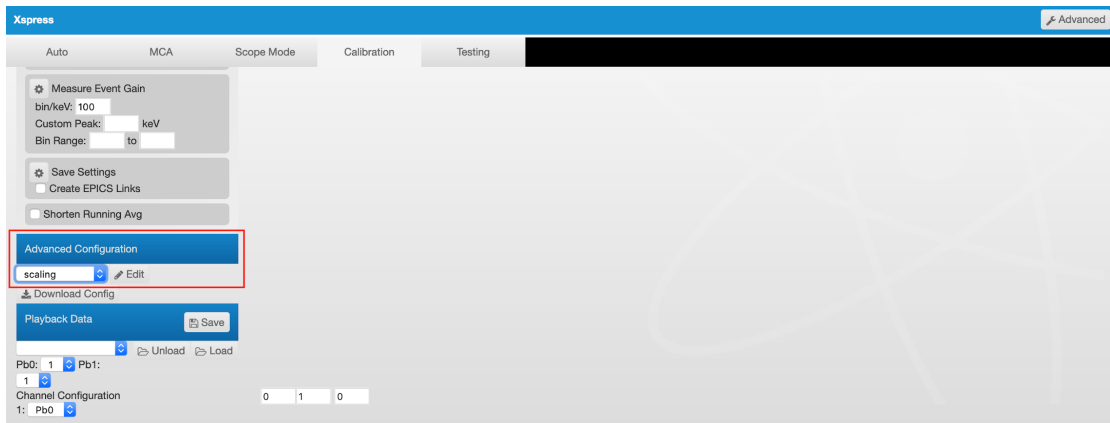
If the scaling calibration fails, or if you are using a target not present in the standard list, then you will need to adjust the calibration manually.

First attempt to run the routine again by typing your target peak energy into the 'Custom Peak' box. Click on the cog. Once complete, click on the 'MCA' tab and click on the 'Collect Data' cog. The peak should now be in the correct place according to the energy calibration.

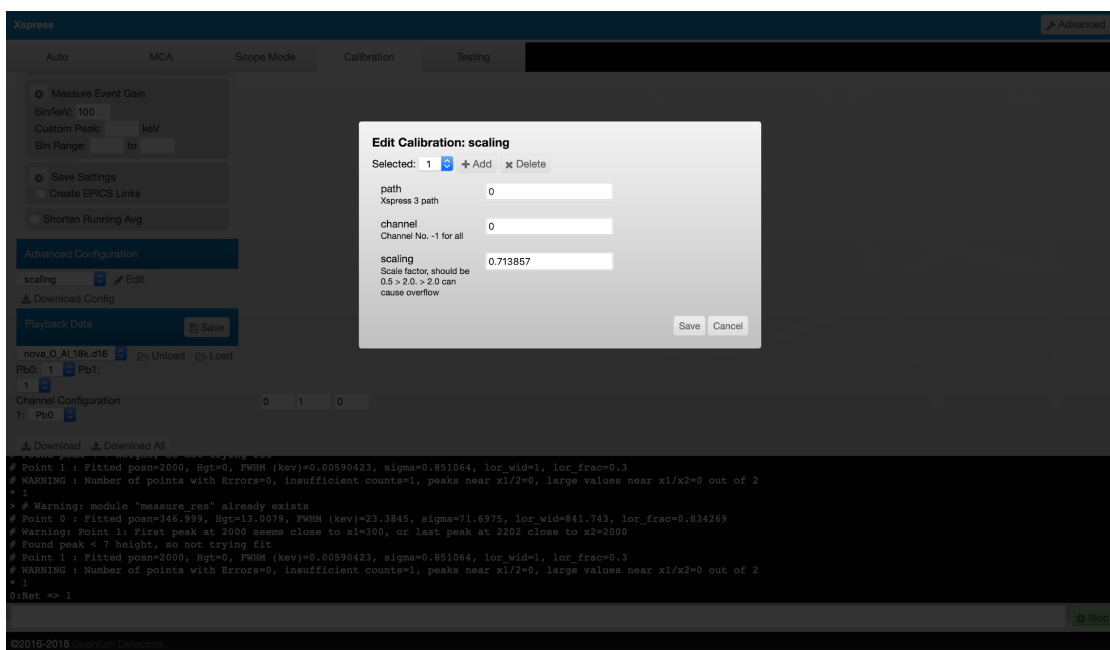
If this fails then the system may be picking out the wrong peak by which to scale the energy. In this case one must manually set the scaling.

Manual scaling

A manual scaling factor can be assigned by editing the scaling value in the Advanced Configuration section, highlighted below in red.



Select scaling from the dropdown list and then click 'Edit'.



You'll be presented with the 'Edit Calibration: scaling' dialogue box, as shown above.

Make a note of the current scaling factor, here 0.713857. Then run an MCA and note the position of your target peak.

The proper scaling factor can be calculated by the equation:

$$\text{Proper_factor} = \text{Current_factor} * \text{Target_peak_position} / \text{Current_peak_position}$$

Enter your calculated factor into the advanced scaling dialogue box and then check by running an MCA. Adjust as necessary.

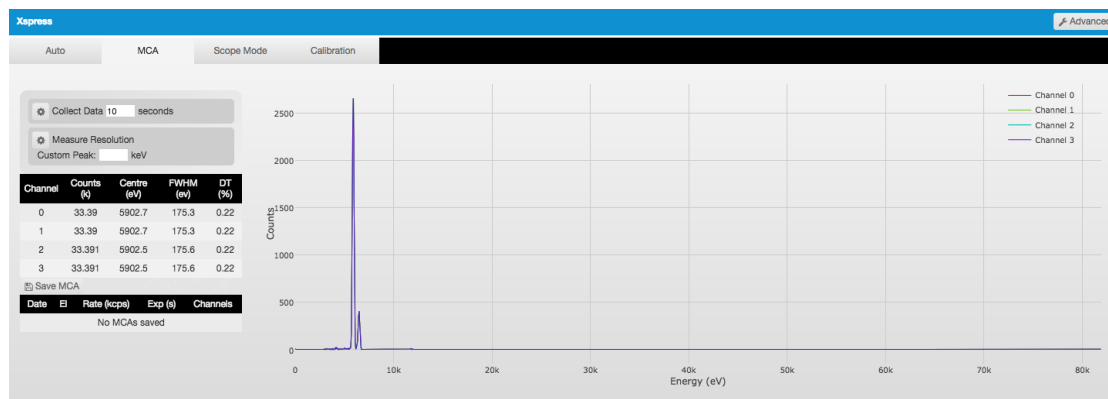
MCA scaling

Users of Germanium detectors may wish to collect spectra up to 200 keV. In the standard configuration, Xpress 3 X assigns 100 bins per keV on its MCA, and there

are a total of 4096 bins. This gives a spectrum from 0-40 keV. To enable spectra at higher energies, one simply has to rescale the MCA to use fewer bins per keV.

Enter your desired bin per keV scaling in the 'bins/keV' box. In the example below, 50 bins per keV gives a spectrum of 0-80 keV:

Click the cog icon next to 'Measure Event Gain' and wait for the process to complete. Then click on the 'MCA' tab and click on the 'Collect Data' cog.



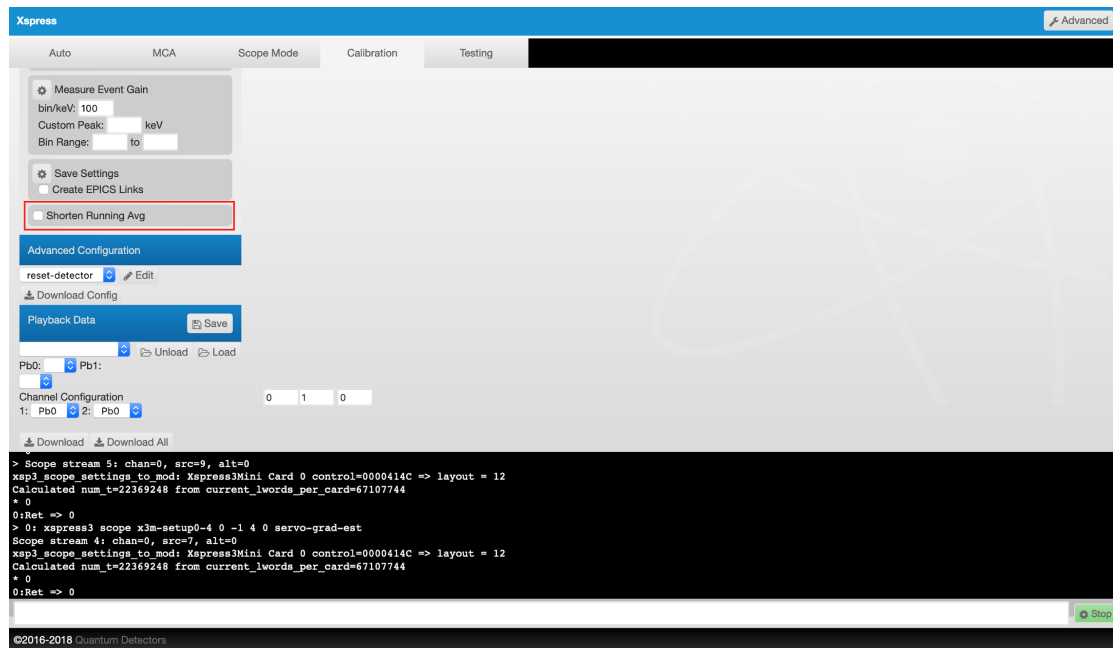
The scale has now shifted such that one could record spectra up to 80 keV.

If your peak is in the wrong position then follow the energy calibration process again.

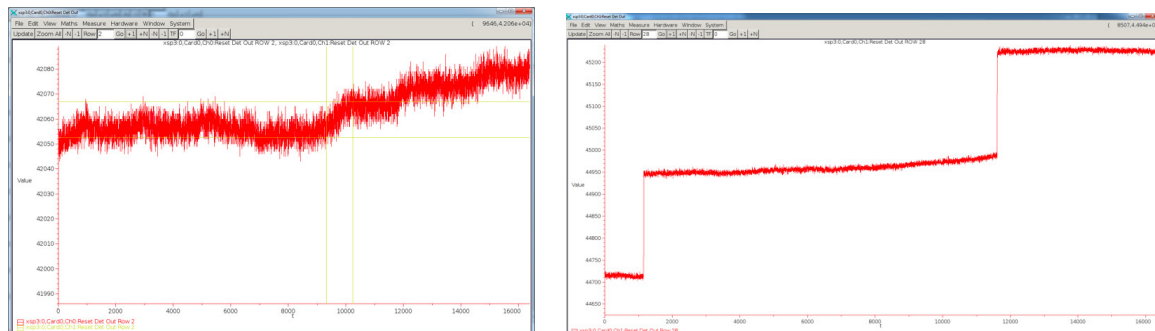
Save Settings

Once the calibration is satisfactorily completed use the 'Save Settings' box to save all the settings to disk.

Shorten Running Average



In some situations, particularly on older detectors, you may experience noise manifesting as a "wobble" in the signal between events. Additionally some of the flat sections between events may not be flat and may slope downwards. Since Xpress 3 needs to average along these "flat" sections, any wobble or variation in gradient can have a detrimental effect on energy resolution. This issue is a particular problem in low flux conditions where there are long periods between events.



Example of a detector exhibiting an inconsistent leakage gradient

To overcome this issue one can activate the 'Shorten Running Avg' option. This will decrease the time over which the flat section of the signal is measured, thus reducing the error in the model caused by the wobbling effect. Take a measurement of the resolution both before and after activating this option. If an improvement is observed then leave it activated, otherwise deactivate it.

5. Multibox Mode (>8 channels)

Multiple Xspress 3 X units can be linked via TTL cables and read into the same PC, this enables the user to see the devices a single readout system with channel numbers from 0-n.

To use a system that has been shipped to you as a multibox system (a 16 channel Xspress 3 X for example), connect the Xspress 3 X Ethernet cables to the host server's 10gb Ethernet card.

One of the Xspress 3 Xs will be 'branded' as Box 0, with other units incrementing upwards from there. You can alter this branding should you need to, this is detailed below. When you receive these from Quantum Detectors, they will be labelled either on the base or increasing serial number order (the lowest serial number will be Box 0)



Box 0 provides TTL signals via the TTL OUT sockets on the front. Connect a cable from TTL OUT 0 to the TTL IN 1 sockets on the next box. You can daisy chain from box to box in this fashion, from TTL OUT 0 to TTL IN 1 each time. This last termination can be set to 50 Ω in software and will be pre-configured ready for use.

To calibrate the multibox version, be certain to first go to Advanced > Calibration and set the number of cards to the number of boxes you have and the number of channels appropriately. Then restart the python program and refresh the browser to ensure this has taken effect. After this, run everything as above for the single box mode.

Changing the 'branding' of box numbers to enable multibox setup

Default IP addresses on the local network between the Xspress 3 Xs and the PC will be 192.168.0.2 for box 0. The 2 needs to increment as the box numbers increase, so the IP for box 1 becomes 192.168.0.3 then 4, 5, ..

All MAC addresses on the gigabit ethernet cards are identical, again, these need to be different and we would suggest that you increment them by 1.

Connect an X3X to a PC via USB, start screen `/dev/ttyUSB0`, reboot X3X and stop uboot via screen (pressing any key)

```
setenv ipaddr <ip addr>
setenv ethaddr <mac addr>
saveenv
reset
```

Power cycle each X3X that has had its box number changed.

All Xspress 3 Xs need connecting to the server directly.

Before you run calibration, please check that there are as many 'cards' in Advanced -> Calibration as there are Xspress 3 Xs.

To change back to single Xspress 3 X use, connect to the X3X again as above and change the IP address back to 192.168.0.2. The MAC address simply needs to be unique in all circumstances, not any particular value.

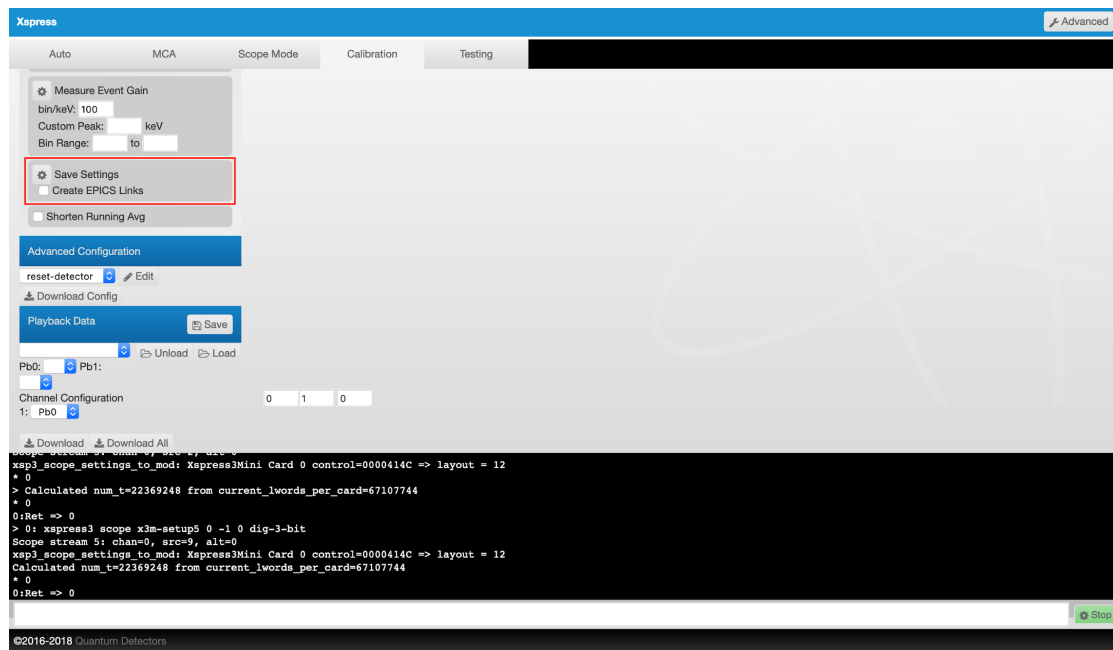
6. System integration

Data collection on Xspress 3 X is usually controlled using either EPICS or TANGO systems.

The EPICS driver for Xspress 3 is based on AreaDetector1.9.

The TANGO driver for Xspress 3 is based on LIMA and TANGO 9.25a.

Once the calibration steps have been completed, ensure that the calibration settings have been fully saved by navigating to the 'Calibration' tab of advanced settings and clicking 'Save Settings'.



Click 'Save Settings' to save all settings from the current calibration.

Close the calibration browser window and make sure the background python script is not running. The python script can be killed by pressing ctrl-c in the terminal that is running it. Copy the settings files to the correct location on your control system.

Starting EPICS

Kill the python calibration script with CTRL-C before running an EPICS IOC.

The EPICS IOC can be started by typing the following command into a terminal:

```
xspress3-ioc.sh
```

The supplied GUI for EPICS is either EDM or MEDM. These screens can be started by opening a new terminal and typing either:

```
xspress3-edm.sh
```

or

```
xspress3-medm.sh
```

If you will be controlling Xspress 3 X using EPICS, copy the settings files from the settings directory to the directory that your EPICS IOC expects to find them in. On our server, with an 8 channel X3X system, one would type the following:

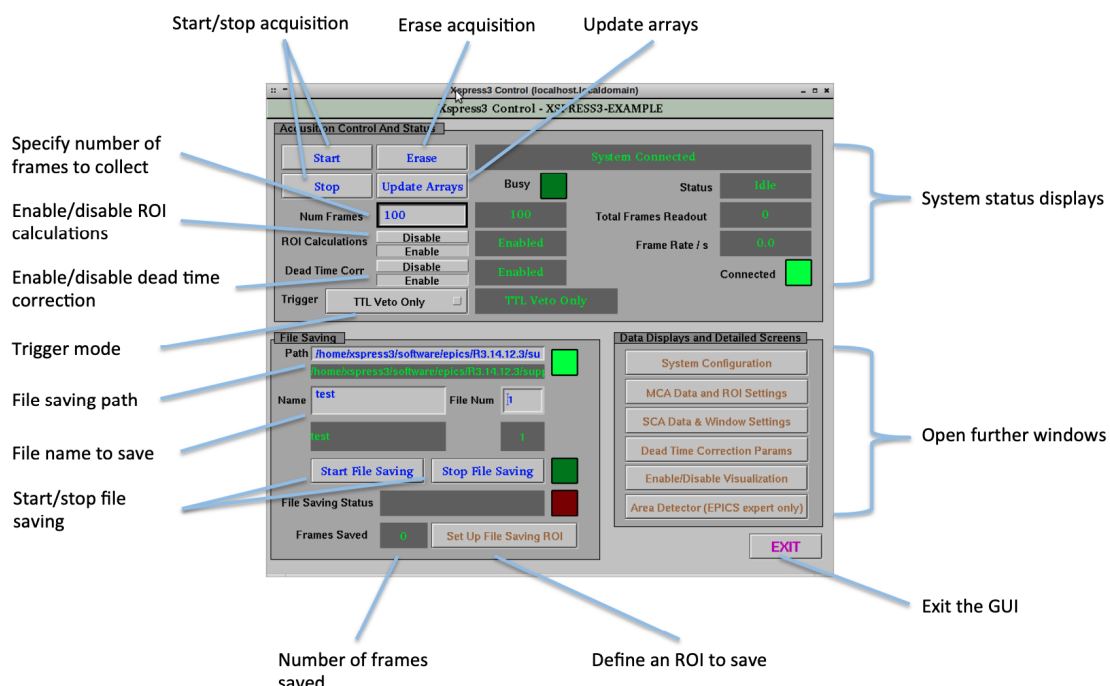
```
sudo cp /etc/xspress3/calibration/initial/settings  
/etc/epics/xspress3/settings/8channel
```

Soft IOC and Xspress 3 control window

When the GUI is started a soft IOC window will open, displaying information on the EPICS version that is in use, host name, user name, the number of channel access clients and the number of channel access PV connections. There is also some information on the machine that the IOC is running from, including OS, memory usage and CPU loading.



Alongside the soft IOC window, the Xspress 3 control window will open.



The Xspress 3 control window is used to control data acquisition. To start a simple acquisition, you should first set up the Trigger mode. *Software*, *Internal* and *TTL Veto Only* are the only modes in use.

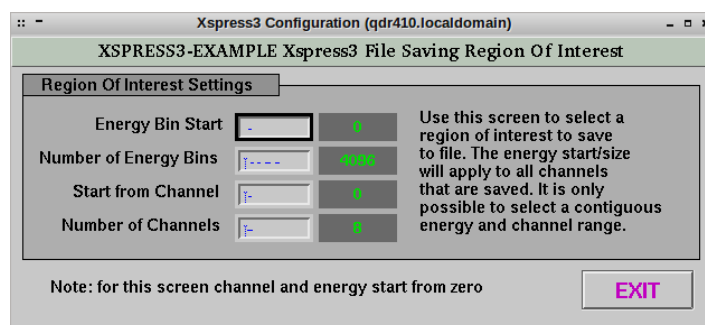
If *Software* mode is selected, Xspress 3 will take a single frame of data when the *Start* button is clicked.

If *Internal* mode is selected, Xspress 3 will take a specified number of frames of given length. Type the number of frames you wish to collect into the *Num Frames* box and press enter, then type the duration of each frame into the *Exp Time* box and press enter.

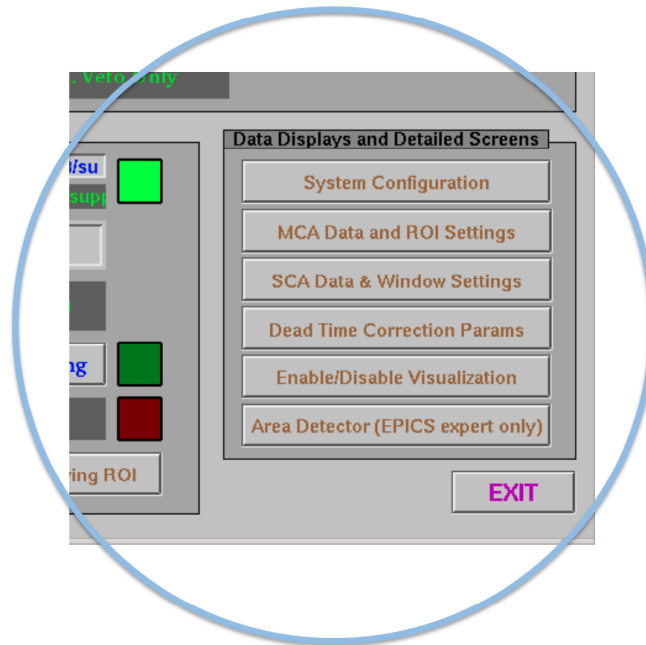
If *TTL Veto Only* is selected, an external trigger must be plugged in to TTL IN 1 on the front of Xspress 3 box 1 in order to trigger the frames. Type the number of frames you wish to collect into the *Num Frames* box and press enter. Click on *Start* to begin collecting data. The frame rate will be determined by the frequency of the TTL signal from your external trigger.

Clicking on *Stop* will stop the current acquisition. Clicking on *Erase* and then *Update Arrays* clears the data from the memory and the MCAs.

If you wish to save data, type a filename into the *Name* box and verify the path to the directory that you want to save to. Clicking on *Start File Saving* before or during an acquisition will save any subsequent frames of data to an HDF5 file. Clicking on *Stop File Saving* during an acquisition prevents any subsequent frames from being saved. If you wish to set up a region of interest to save, click on *Set Up File Saving ROI*. The dialog box pictured below will open, allowing you to specify the channels and energy range that you are interested in saving data from.

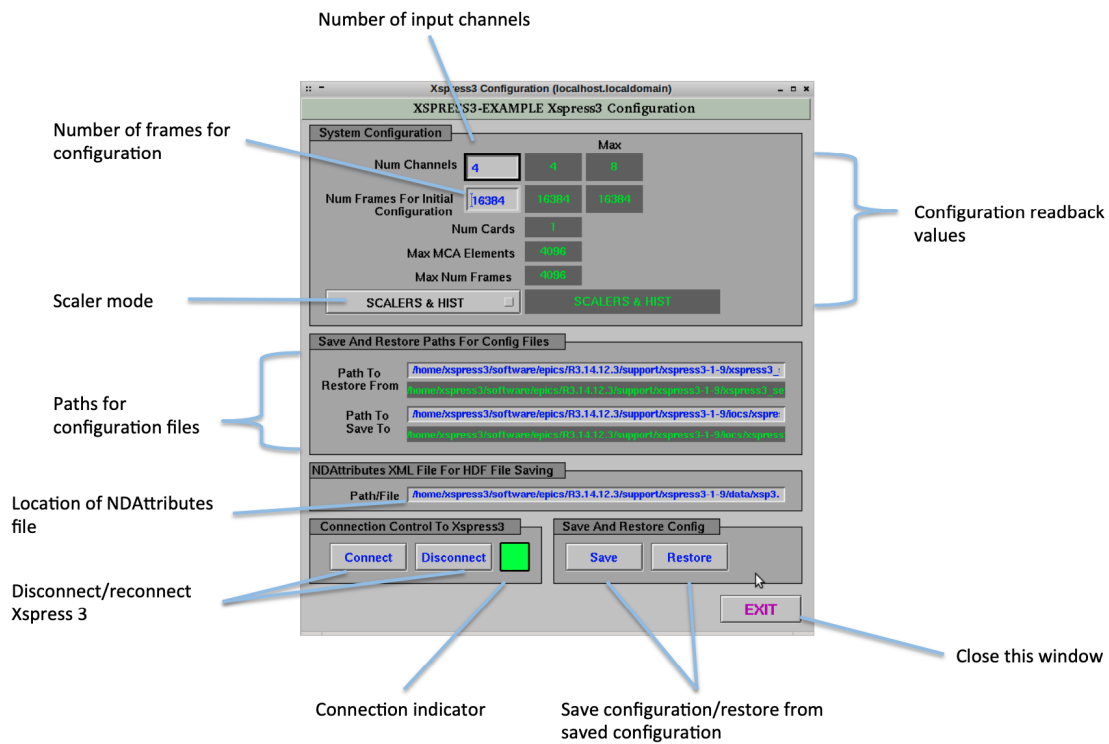


Further windows may be opened by clicking on the buttons at the lower right side of the control window.



System configuration

Clicking on the *System Configuration* button opens the System Configuration window.



In this window you can specify the number of input channels that Xspress 3 uses, specify the location of your Xspress 3 configuration files, save a new configuration or restore from a saved configuration.

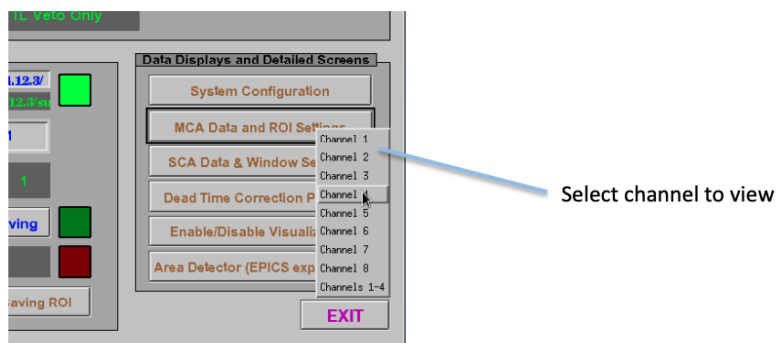
Clicking on the *Scalers & Hist* button lets you select between *Scalers & Hist* and *Playback* modes. In normal operation, *Scalers & Hist* should be selected.

By default, the configuration files created during calibration are located in the *Path To Restore From* box. The *Path To Save To* box points to a different directory so that the original configuration files are not accidentally overwritten.

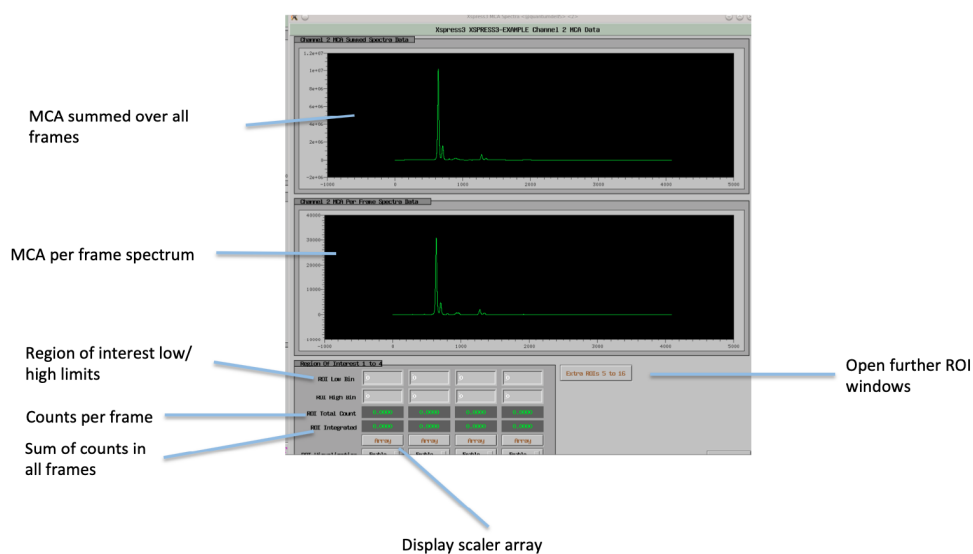
The configuration readback values show the current configuration. In the example above, 4 out of a possible 8 input channels are in use. A single card is in use (i.e. a single Xspress 3 unit), and the maximum number of MCA elements and frames is 4096. The upper limits for each of these values are set by firmware, and the user is not able to exceed them.

MCA data and region of interest settings

Clicking on the *MCA Data and ROI Settings* button brings up a menu from which the MCA for a particular channel may be selected.

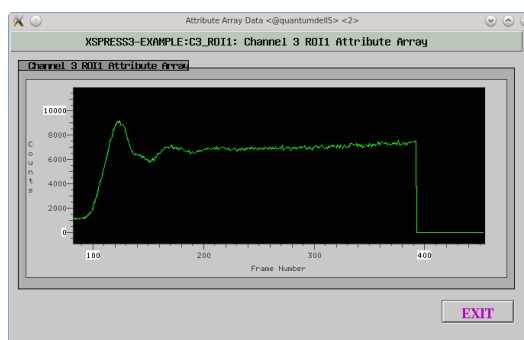


The MCA window contains two panes showing MCA traces. The top pane shows MCA data summed over all frames, whilst the bottom pane shows MCA data per frame. The traces update in real time as the data is collected, allowing you to view your acquisition as it progresses.



Regions of interest may be set by entering the high and then low limits for the region in which you are interested.

To view a scaler array for a region of interest, click on the *Array* button. The example below shows an EXAFS scan of copper.



SCA data and window settings

Clicking on the *SCA Data and Window Settings* button allows you to define windows for the collection of events in a specified energy range.

Events in user-defined window

Elapsed time (ns)

Number of detector resets

Total events

Total good events

Pileup

	Count	Low	High
Window 1	0.0000	0	0
Window 2	0.0000	0	0
Time	199907.0000		
Reset Ticks	972862.0000		
Reset Count	1000.0000		
All Event	410151.0230		
All Good	385196.0000		
Pileup	147.0000		

Threshold: 0

EXIT

If no window is specified, values for the elapsed time (ns), detector resets and events across the entire MCA range will be displayed.

Dead time correction

Clicking on the *Dead Time Correction Params* button displays the dead time correction parameters in a new window. In the example below, the system is at idle so no dead time correction is being applied.

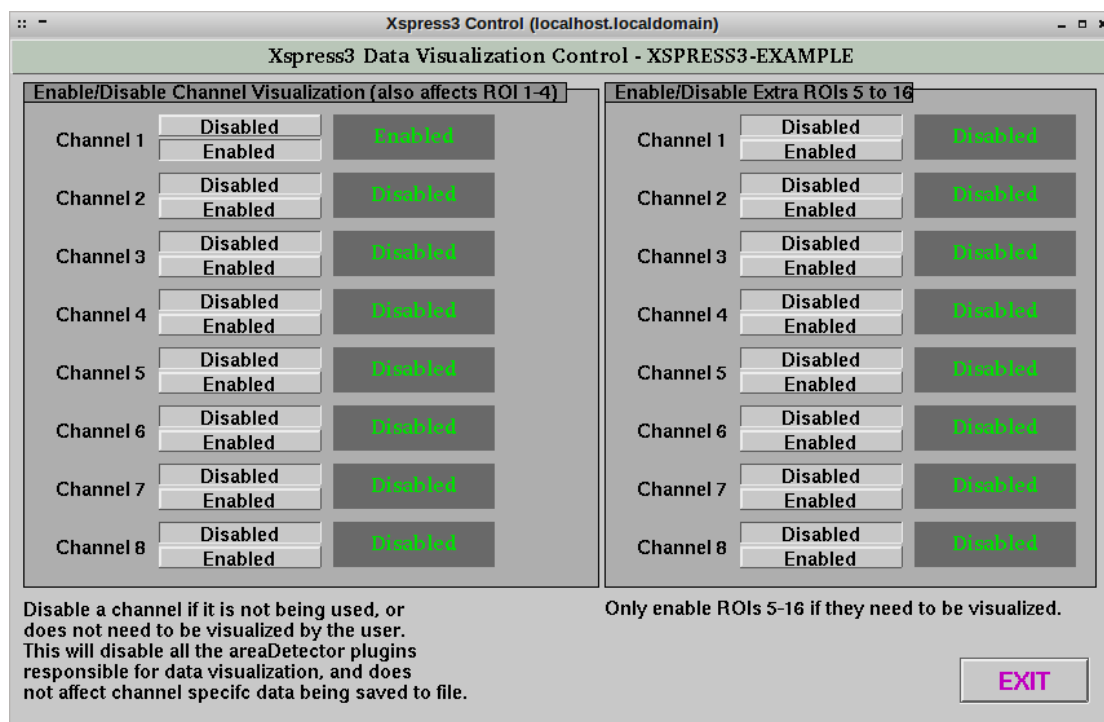
Channel 1 Dead Time Correction Parameters

Flags	0
All Good Event Gradient	0.0000
All Good Event Offset	0.0000
In Window Gradient	0.0000
In Window Offset	0.0000

EXIT

Enable/disable visualization

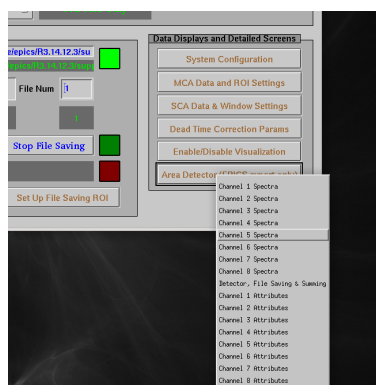
Clicking on the *Enable/Disable Visualization* button brings up a new window which allows you to choose which channels are visible to the user.



Channels must be enabled using this window before the user is able to view them using the MCA windows. Enabling or disabling a channel here does not affect the content of any HDF5 files that will be saved. Those files will contain data from all channels regardless of whether or not they have been enabled here.

Area detector

The Area Detector window is only for use by EPICS experts, and so will not be described in detail here. The window is opened as shown in the example below:



7. EPICS IOC

This EPICS module provides areaDetector based control of the Xspress3 Readout System. It uses an API provided by Quantum Detectors and STFC to control the hardware and readout data.

This section provides instructions on how to use the EPICS software interface to Xspress 3 if you'd prefer not to use our GUI. PV (process variable) information is also available by middle clicking on any element of the GUI.

This software is licensed under the GNU Lesser General Public License.

Xspress 3 IOC process variables

Once the EPICS IOC is running it is necessary to connect to the Xspress 3 system. This can be done by setting the following PVs. The {P} is the base part of the PV name, which is system dependent. PVs with an _RBV on the end are readback PVs that can be used to confirm that a parameter has successfully been changed.

{P}:NUM_CHANNELS	The number of channels this system is configured for. Must be less than or equal to {P}:MAX_NUM_CHANNELS_RBV, which is the value passed into xspress3Config (a function used at IOC initialisation by the IOC developer).
{P}:NUM_FRAMES_CONFIG	The number of frames to configure the system with. This must be less than or equal to {P}:MAX_FRAMES_RBV, which is the value passed into xspress3Config. Normally this should be set equal to {P}:MAX_FRAMES_RBV.
{P}:CONFIG_PATH	This is the path to the Xspress 3 configuration files. These files should have been provided with the system.
{P}:NDAttributesFile	This is the path to the Xspress 3 XML file saving configuration files. This file should have been provided with the IOC software. It defines what parameters are saved in the HDF file and what can be visualized when running the system. An example that can be used is packaged in the IOC data directory.

Once the above PVs are set, connection can take place by writing 1 to {P}:CONNECT. If connection was successful then {P}:CONNECTED will be set to 1. EPICS ca_put_callback is supported by the {P}:CONNECT PV.

When connection occurs the following API functions are called by the EPICS driver:

1. xsp3_config
2. xsp3_clocks_setup
3. xsp3_restore_settings
4. xsp3_format_run
5. xsp3_set_run_flags

In addition, if connection was successful, several parameters are read from the system so the EPICS driver can update RBV type PVs (e.g. scaler window limits).

Data Acquisition

The following PVs are involved in data acquisition or monitoring. Some of the PVs map directly onto Xspress 3 API functions, and this will be noted below.

The basic flow of control, after connection, will be:

1. Set trigger mode (Select 'TTL Veto Only' for external triggering, where a LVTTTL high signal will enable counting and the rising and falling edge will define the frame start and end points).
2. Set up file related PVs - FilePath, FileName. Check FilePath exists.
3. Set file number, if needed. This is the index used in the filename. This is auto incrementing each time a new file is saved, but is reset if the IOC is restarted.
4. If the IOC has just been restarted it will be necessary to take a single frame to configure the file writing areaDetector plugin:
 1. Set number of frames to 1.
 2. Start acquisition and generate a single external trigger.
 3. Wait until acquisition has finished.

Then for each acquisition:

1. Erase the data.
2. Set the number of frames for the real data collection.
3. Start file saving.
4. Start acquisition.
5. Wait until acquisition has finished, then check status and number of frames processed.
6. Update the visualization arrays before reading the array data over channel access (also do this during acquisition before reading arrays).
7. Wait until file saving has finished, then check status and number of frames saved to disk.

The user may also wish to set up regions of interest (ROIs) (see section below), which must be done before starting an acquisition.

{P}:Acquire {P}:Acquire_RBV	This is used to start and stop acquisition. Write a 1 to this to begin acquisition. Normally the data collection will finish automatically once NumImages is reached. However, to manually stop data collection write a 0 to this PV. If the driver is in the process of reading a large number of frames, for example at high frame rates, it will finish this before the stop completes. This PV maps directly to xsp3_histogram_start and xsp3_histogram_stop . When writing a 1 to {P}:Acquire, ca_put_callback is supported. NOTE: it can take about 2s after starting acquisition before the system is ready to accept triggers.
{P}:ERASE	This will call the Xspress 3 API xsp3_histogram_clear function. It will also clear any data held by EPICS PVs, including the array PVs. Do this before starting a new data collection.
{P}:UPDATE	This does a manual update on all the scaler and ROI arrays, to ensure that they contain the latest data. It is necessary to

	write a 1 to this at the end of an acquisition, or during acquisition, if reading the data over channel access.
{P}:NumImages {P}:NumImages_RBV	Set and read the number of frames to acquire. This number must be less than {P}:NUM_FRAMES_DRIVER_RBV, which was set when the IOC was configured by the developer. Acquisition will automatically stop when the number of frames acquired equals NumImages_RBV.
{P}:CTRL_MCA_ROI {P}:CTRL_MCA_ROI_RBV	This is used to enable and disable ROI calculations. To enable ROI calculations set this to 1 and set the ROI limits. To turn ROI calculations off set this to 0, which can save CPU time if operating at high frame rates and ROIs are not needed.
{P}:CTRL_DTC {P}:CTRL_DTC_RBV	This is used to determine if dead time corrected data is read out. If this PV is enabled then the MCA spectra will be dead time corrected, using the API function xsp3_hist_dtc_read4d . If this PV is disabled then un-corrected MCA spectra data is read out using xsp3_histogram_read4d . This option does not affect scaler data, which is always dead time corrected. The dead time corrected parameters are saved in the HDF file (see below).
{P}:TriggerMode {P}:TriggerMode_RBV	Set and read the triggering mode. This PV maps directly on to the API function xsp3_set_glob_timeA . In simulation mode any mode can be selected. Options are: • Fixed - Corresponds to API option XSP3_GTIMA_SRC_FIXED • Internal - Corresponds to API option XSP3_GTIMA_SRC_INTERNAL • IDC - Corresponds to API option XSP3_GTIMA_SRC_IDC • TTL Veto Only - Use this for external triggering using a gate signal. Corresponds to API option XSP3_GTIMA_SRC_TTL_VETO_ONLY • TTL Both - Corresponds to API option XSP3_GTIMA_SRC_TTL_BOTH • LVDS Veto Only - Corresponds to API option XSP3_GTIMA_SRC_LVDS_VETO_ONLY • LVDS Both - Corresponds to API option XSP3_GTIMA_SRC_LVDS_BOTH
{P}:StatusMessage_RBV	This contains a string to describe the current state of the EPICS driver. Any error should result in a meaningful message in this PV.
{P}:DetectorState_RBV	This PV contains the current state of the EPICS driver. See the areaDetector documentation for a list of the possible states. Most of the time the state will be either Idle or Acquire . In the event of an error it will be Error .

{P}:FRAME_COUNT_RBV	This contains the number of frames read out of the Xspress 3 each time the driver checks for new data. Only really useful for debugging.
{P}:ArrayCounter_RBV	Contains the total number of frames acquired during the current or last acquisition. It is reset to zero at the beginning of the next acquisition. The rate of change of this PV also indicates the frame rate we are operating at.
{P}:ArrayRate_RBV	This is a rough indication of the frame rate we are operating at. It is calculated every second, and uses the rate of change of the ArrayCounter_RBV PV.

HDF File Saving

This software makes use of the areaDetector HDF file writing plugin. Full documentation for this is provided at:

<http://cars9.uchicago.edu/software/epics/NDPluginFile.html>.

Some of the most relevant PVs are exposed on the main GUI, and these are detailed below:

{P}:HDF5:Capture {P}:HDF5:Capture_RBV	Write a 1 to this PV before starting an acquisition. This will trigger the HDF software to start capturing data to write a file, and set {P}:HDF5:Capture_RBV to 1. At the end of the acquisition it is necessary to write a 0 to this PV, unless {P}:HDF5:Capture_RBV has already been set to zero because the required number of frames have been saved.
{P}:HDF5:FilePath {P}:HDF5:FilePath_RBV	The file system path to use for file saving. If the path exists then {P}:HDF5:FilePathExists_RBV will go to 1.
{P}:HDF5:FilePathExists_RBV	A flag indicating if the set file path is valid.
{P}:HDF5:FileName {P}:HDF5:FileName_RBV	The file name to use.
{P}:HDF5:FileNumber {P}:HDF5:FileNumber_RBV	This is the index of the next file that will be saved. If using EPICS autosave this will be reset to the last value when the IOC is rebooted.
{P}:HDF5:WriteStatus	The status of the file writing. 1=Error. This should be checked at the end of each acquisition, once file saving has completed.
{P}:HDF5:WriteMessage	The error message if {P}:HDF5:WriteStatus is 1.
{P}:HDF5:NumCaptured_RBV	This is the number of files saved to disk by the areaDetector HDF plugin. At the end of file saving this should equal the number of frames in {P}:ArrayCounter_RBV.

Saving a region of interest (ROI) to disk

In order to save a ROI of data to disk the data is sent through an Epics areaDetector ROI plugin. Full documentation is at:

<http://cars9.uchicago.edu/software/epics/NDPluginROI.html>

However, the average user need only use the following PVs. For each frame, the data is arranged in an X*Y array. X is energy and Y is channel number. The user can set the

start and end point in terms of energy, and which channels to save. It is only possible to save a contiguous range of energy or channels.

These PVs have to be set before the beginning of an acquisition.

{P}:ROIDATA:MinX {P}:ROIDATA:MinX_RBV	The start point in units of energy bin (0 based)
{P}:ROIDATA:SizeX {P}:ROIDATA:SizeX_RBV	The size of the energy ROI to save.
{P}:ROIDATA:MinY {P}:ROIDATA:MinY_RBV	The start point in channel number (0 based, so channel 1 is 0, and channel 8 is 7).
{P}:ROIDATA:SizeY {P}:ROIDATA:SizeY_RBV	The number of channels to save.

By restricting the energy range that is saved to disk, and the number of channels, the file size can be significantly reduced.

Scaler Data and Window Settings

The following PVs provide access to the scalers over channel access, and allow the user to set scaler related parameters. The 'C1' part of the PV name indicates channel 1. The lists is duplicated for each channel: 'C2', 'C3' etc.

The first table shows the single value PVs that show the latest scaler value during an acquisition.

{P}:C1_SCA0:Value_RBV	The Time scaler.
{P}:C1_SCA1:Value_RBV	The Reset Ticks scaler.
{P}:C1_SCA2:Value_RBV	The Reset Count scaler.
{P}:C1_SCA3:Value_RBV	The All Event scaler.
{P}:C1_SCA4:Value_RBV	The All Good scaler.
{P}:C1_SCA5:Value_RBV	The Scaler 1 scaler. The lower and upper window limits for this are set using {P}:C1_SCA5_LLM and {P}:C1_SCA5_HLM. The window limits are set using API function xsp3_set_window
{P}:C1_SCA6:Value_RBV	The Scaler 2 scaler. The lower and upper window limits for this are set using {P}:C1_SCA6_LLM and {P}:C1_SCA6_HLM. The window limits are set using API function xsp3_set_window
{P}:C1_SCA7:Value_RBV	The Pileup scaler.

The following table shows the set and read parameters for the **All Good**, **Scaler 1** and **Scaler 2** scalers.

{P}:C1_SCA4_THRESHOLD {P}:C1_SCA4_THRESHOLD_RBV	Set and read the All Good scaler window threshold. This is set using the API function xsp3_set_good_thres The readback value is also read at connection time.
{P}:C1_SCA5_LLM {P}:C1_SCA5_LLM_RBV	Set and read the Scaler 1 scaler window lower limit. This is set using the API function xsp3_set_window . The readback value is also read at connection time.
{P}:C1_SCA5_HLM {P}:C1_SCA5_HLM_RBV	Set and read the Scaler 1 scaler window upper limit. This is set using the API function xsp3_set_window The readback value is also read at connection time.

{P}:C1_SCA6_HLM {P}:C1_SCA6_LLM_RBV	Set and read the Scaler 2 scaler window lower limit. This is set using the API function xsp3_set_window The readback value is also read at connection time.
{P}:C1_SCA6_HLM {P}:C1_SCA6_HLM_RBV	Set and read the Scaler 2 scaler window upper limit. This is set using the API function xsp3_set_window The readback value is also read at connection time.

The following table shows the scaler data array PVs. These arrays accumulate the scaler data during an acquisition to provide live feedback if needed. At the end of the acquisition the arrays are updated with the final scaler data, providing {P}:UPDATE has been triggered.

{P}:C1_SCA0:ArrayData_RBV	The Time scaler array.
{P}:C1_SCA1:ArrayData_RBV	The Reset Ticks scaler array.
{P}:C1_SCA2:ArrayData_RBV	The Reset Count scaler array.
{P}:C1_SCA3:ArrayData_RBV	The All Event scaler array.
{P}:C1_SCA4:ArrayData_RBV	The All Good scaler array.
{P}:C1_SCA5:ArrayData_RBV	The Scaler 1 scaler array.
{P}:C1_SCA6:ArrayData_RBV	The Scaler 2 scaler array.
{P}:C1_SCA7:ArrayData_RBV	The Pileup scaler.

The above parameters are saved in the HDF file.

MCA Spectra and ROI Data

The following table lists the array PVs for the MCA spectra, as well as the ROI window settings and readback values. There are 16 regions of interest per channel, so only the PVs for the first one are listed in the table below. The ROI PVs below have 'ROI1' in the name, so simply substitute this for 'ROI2', 'ROI3' etc. for the other regions of interest.

The ROI limit PVs have to set before the beginning of an acquisition.

{P}:ARR1:ArrayData	The MCA spectra data (dead time corrected, or un-corrected, depending on {P}:CTRL_DTC. This is updated during a scan when a new frame is read out.
{P}:ARRSUM1:ArrayData	The cumulative MCA spectra data. This is updated during a scan when a new frame is read out. This is the sum of all the spectra data from {P}:ARR1:ArrayData.
{P}:C1_MCA_ROI1_LLM	The lower limit for the first ROI.
{P}:C1_MCA_ROI1_HLM	The upper limit for the first ROI.
{P}:C1_ROI1:Value_RBV	The ROI readback value for each frame. This is updated during acquisition.
{P}:C1_ROI1:ValueSum_RBV	The integrated ROI readback value. This is updated during acquisition.
{P}:C1_ROI1:ArrayData_RBV	The ROI readback array that holds the ROI for every frame from an acquisition. This is updated during acquisition. The PV {P}:C1_ROI1:ValueSum_RBV will hold the integral of the array.

{P}:C1_ROI1:EnableCallbacks	This can be used to enable or disable visualization of the ROI data during acquisition. Other PVs are provided that allow the user to globally disable all ROIs per channel, so this is only available for fine control if needed. By default only ROIs 1-4 are enabled by the high level plugin control PVs.
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The above parameters are saved in the HDF file. All the MCA spectra arrays are also saved.

The Dead Time Correction Parameters

When the Epics IOC starts up it reads the dead time correction (DTC) parameters for each channel. These are set by the system configuration files supplied with the readout system.

{P}:C1_DTC_FLAGS_RBV	The DTC Flags parameter.
{P}:C1_DTC_AEG_RBV	The DTC All Good Event Gradient parameter.
{P}:C1_DTC_AEO_RBV	The DTC All Good Event Offset parameter.
{P}:C1_DTC_IWG_RBV	The DTC In Window Gradient parameter.
{P}:C1_DTC_IWO_RBV	The DTC In Window Offset parameter.

The above parameters are saved in the HDF file.

PVs used to enable and disable data visualization during acquisition

There are several PVs provided that allow the user to enable and disable data visualization. This does not affect file saving or ROI calculation, only the online display of the data during an acquisition. This is provided to allow the user to only enable what is needed, as it can be CPU intensive to visualize the data at high frame rates. It has been found that by having too many channels enabled for visualization, or too many ROIs, the data readout rate can drop. For full speed acquisition, turn off all data visualization, or only enable a single channel.

The following PVs are available for each channel. Substitute the number in 'C1' for the other channel numbers (1-25).

These PVs have to set before the beginning of an acquisition.

{P}:C1_PluginControlVal	This controls the areaDetector plugins responsible for data visualization for channel 1. This includes the spectra, scaler and ROIs 1-4 only. ROIs 5-16 are not affected by this PV. To control ROIs 5-16 use {P}:C1_PluginControlValExtraROI instead.
{P}:C1_PluginControlValExtraROI	This controls the areaDetector plugins responsible for data visualization for the ROIs 5-16. These are not always used, which is why they are separated from the PV {P}:C1_PluginControlVal. It would not normally be desirable to automatically enable ROIs 5-16 when a channel is enabled.